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Peripheral anatomy and central connectivity of proprioceptive sensory neurons in the *Drosophila* wing

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eLife Assessment

This **important** work describes wing mechanosensory neurons in detail, extending our understanding of sensorimotor processing in the fruit fly. The evidence presented **convincingly** supports the authors' identification of these neurons and leverages state-of-the-art methods to generate a near-complete map of wing mechanosensory circuitry. Overall, this study provides new hypotheses and invaluable tools for investigating proprioceptive motor control of the wing in *Drosophila*.

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Abstract

Recent advances in electron microscopy (EM) and automated image segmentation have produced synaptic wiring diagrams of the *Drosophila* central nervous system. A limitation of existing fly connectome datasets is that most sensory neurons are excised during sample preparation, creating a gap between the central and peripheral nervous systems. Here, we bridge this gap by reconstructing wing sensory axons from the Female Adult Nerve Cord (FANC) EM dataset and mapping them to peripheral sensory structures using genetic tools and light microscopy. We confirm the location and identity of known wing mechanosensory neurons and identify previously uncharacterized axons, including a novel population of putative proprioceptors that make monosynaptic connections onto wing steering motor neurons. We also find that proprioceptors of adjacent campaniform sensilla on the wing have distinct axon morphologies and postsynaptic partners, suggesting a high degree of specialization in axon pathfinding and synaptic partner matching. The peripheral location and central projections of wing sensory neurons are stereotyped across flies, allowing this wing proprioceptor atlas and genetic toolkit to guide analysis of other fly connectome datasets.

Introduction

Fly wings are exquisite, versatile biological actuators. During flight, they sweep back and forth through the air hundreds of times per second to keep the fly aloft. On the ground, flies extend their wings to groom and males vibrate a wing to attract females during courtship. To accomplish these myriad functions, wing motor control relies on temporally and spatially precise feedback from diverse sensory neurons distributed throughout the wing (Figure 1A). Proprioceptive mechanosensory neurons play a particularly important role in flight control (Pringle, 1957), as mechanosensory feedback has a shorter latency than visual signals and can therefore be used to rapidly adjust wing motion (Dickerson, 2020). Wings experience dynamic forces during flight, and proprioceptors encode features of these forces such as wing bending, twisting, and load (Dickerson et al., 2021). *Drosophila* typically beat their wings at 200–250 Hz and can adjust wing kinematics from one stroke to the next (Dickinson et al., 1993; Heide and Götz, 1996). Thus, muscle contraction must be temporally precise enough to act at these short time scales (Dickinson and Tu, 1997). Consistent with the need for rapid feedback, some motor neurons that control

wing steering muscles receive monosynaptic input from wing sensory neurons (Fayyazuddin and Dickinson, 1999). However, the peripheral location and identity of the wing sensory neurons that provide feedback to the wing motor system remain largely unknown.

Sensory neurons on the *Drosophila* wing can be grouped into different classes based on their end organ morphology (Figure 1B). The most numerous are the bristles along the wing margin, which include both mechanosensory and chemosensory sensilla (Hartenstein and Posakony, 1989; Palka et al., 1979). Wing chemosensory neurons can detect external odors and pheromones (Stocker, 1994), while mechanosensory bristles can detect the presence of dust particles or mites (Hampel et al., 2017; Li et al., 2016). Bristles also line the tegula, a cuticular protuberance at the proximal edge of the wing. Apart from the tactile and chemosensory bristles, other sensory neuron classes are presumed to be proprioceptive, in that they monitor the movement and strain of the wing itself. These include campaniform sensilla (CS) and chordotonal organs (CO), and the hair plates (HP), all of which can also be at other locations across the adult fly body, including the legs (Dinges et al., 2021; Field and Matheson, 1998). CS consist of a single neuron with a dendrite that contacts a cuticular cap, or dome, on the surface of the wing; the CS neuron fires action potentials when the dome deforms (Chapman et al., 1973; Moran et al., 1971; Pringle, 1938a). CS can be found as single CS or in fields of domes that have similar size and orientation (Cole and Palka, 1982; Dinges et al., 2021). COs are clusters of neurons with cap cells that anchor them to an internal structure, such as a tendon (Field and Matheson, 1998). In the wing, they are anchored to inner extensions of the cuticles; for example, the wall of the tegula and the inner wall of the radius (sometimes called the radial vein). Hair plates are small tightly packed clusters of sensory hairs, each of which is innervated by a single mechanosensory neuron (Pringle, 1938b). Proprioceptive neurons (CS, CO, and HP) are concentrated proximally, especially along the radius and the tegula (Figure 1C). The axons of wing sensory neurons project into the fly's ventral nerve cord (VNC), the invertebrate analog of the spinal cord. Previous work has described the activity of fly leg proprioceptors during walking (Dallmann et al., 2024; Pratt et al., 2025), but it has been prohibitively challenging to record activity of wing sensory neurons during flight.

Much of what we know about wing sensory neurons comes from developmental studies that used the fly wing as a model to investigate whether axonal morphology is intrinsically determined or extrinsically directed. Some studies used mosaic mutant flies with hindwings in place of halteres to test whether sensory axons would follow haltere-like morphologies or wing-like morphologies once they entered the developing central nervous system (Ghysen, 1978; Palka et al., 1979). These studies measured morphological similarities between wild-type and mutant axons to uncover their intrinsic developmental programs. Their findings showed that the degree of intrinsic programming was different for single CS and field CS, in that axons from field CS on the mutant hindwings followed similar paths in the VNC to the field CS on wild type halteres, while the axons of single CS on mutant hindwings retained the morphological characteristics of the wild type forewing single CS axons (Palka et al., 1979). This difference suggests that the field and single CS are endowed with different axon guidance instructions, connect to different postsynaptic partners, and thus may serve distinct functions.

Understanding how central circuits integrate information from wing sensory neurons is key to understanding their function. Connectomics, or dense reconstruction of neurons and synapses from electron microscopy, offers new opportunities for mapping peripheral sensory feedback to the CNS (Galili et al., 2022). In this study, we bridge the gap between the VNC connectome and the wing by mapping central axon morphologies to the peripheral structures from which they originate (Meinertzhagen et al., 2009). We reconstructed all 490 afferents in the left wing nerve (Anterior Dorsal Mesothoracic nerve, ADMN) in the FANC electron microscopy dataset (Azevedo et al., 2024; Phelps et al., 2021). Many axon morphologies and their corresponding peripheral end organs were previously undescribed. We identified genetic driver lines for a subset of these unknown wing sensory neurons and elucidated their peripheral location and anatomy. For example, we identified novel classes of peripheral sensory neurons near the wing hinge and found that campaniform sensilla on the tegula synapse directly onto the tonic wing b1 motor neuron,

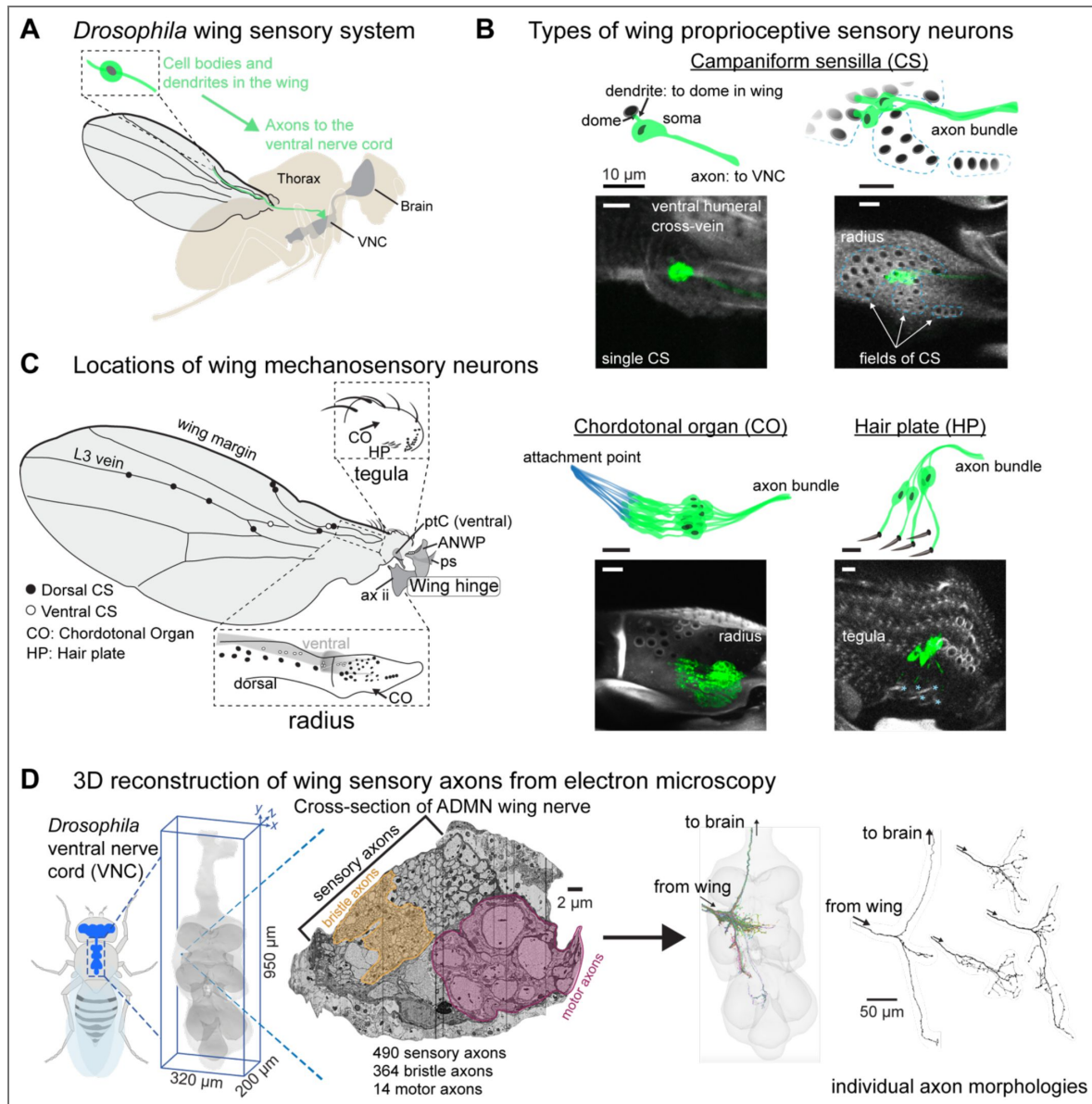


Figure 1. Proprioceptive neurons on the *Drosophila* wing.

(A) The cell bodies and dendrites of sensory neurons are in the periphery, on the wing and wing hinge, and their axons project to the ventral nerve cord (VNC). Before entering the VNC, the sensory axons fasciculate together and enter through the Anterior Dorsal Mesothoracic Nerve (ADMN). (B) Proprioceptors on the wing include campaniform sensilla (CS), chordotonal organs (CO), and a hair plate (HP). Each campaniform sensillum dome is innervated by a single sensory neuron, as is each hair in a HP. A CO is made up of a group of sensory neurons with supporting cells that fix them to the underside of the cuticle (shown in blue). Blue asterisks (bottom right) indicate a single external hair in the HP. Images show the membrane-bound fluorescent label mCD8::GFP to highlight each proprioceptor type. See Table 3 for details on which proprioceptors are labeled by which driver lines, driver lines for representative images in this panel are: single CS (12C07-GAL4); field CS (10G03-GAL4); CO (15F10-GAL4); HP (16C09-GAL4). Scale bars are 10 μ m. (C) Location of sensory neurons on the wing and wing hinge. The location of sensory neurons and the number of CS in each field are based on confocal images and a prior study (Dinges et al., 2021). A subset of sclerites and other structures that make up the wing hinge are included as landmarks: pterale C (ptC), the anterior nodal wing process (ANWP, which also features three CS), the parascutal shelf (ps), and the second axillary (ax ii). (D) We reconstructed each sensory axon in the ADMN wing nerve to visualize its full morphology and analyze downstream connectivity in the VNC. More information on each of these steps in **Methods** and Azevedo et al., 2024. In the nerve cross-section, the motor domain and margin bristle domains are highlighted by outlined gray masks.

suggesting a specialized role in feedback control of flight steering. We also confirmed a long-standing prediction that individual campaniform sensilla from the same field can have distinct axon morphologies (Palka et al., 1986 [DOI](#)). A companion paper that reconstructed haltere campaniform sensilla axons in the connectome identified a similar organization (Dhawan et al., 2025 [DOI](#)). Overall, knowing the relationships between peripheral neuroanatomy, axon morphology, and downstream connectivity to wing motor neurons provides a foundation for investigating proprioceptive sensing and motor control of the fly wing.

Results

Comprehensive reconstruction of wing axons in the FANC connectome

We reconstructed all axons in the left ADMN using an EM dataset of the ventral nerve cord (VNC) of a female adult fly (FANC; Figure 1D [DOI](#); also see **Methods**; (Azevedo et al., 2024 [DOI](#); Phelps et al., 2021 [DOI](#))). For each automatically segmented neuron, we used the software interface Neuroglancer to manually proofread (**Methods**) the major branches, as well as all branches that could be reliably attached. In the left ADMN, we identified 490 sensory axons and 14 motor axons. Axons were identified as sensory if they did not attach to a cell body in the VNC (**Methods**). The total number of axons is slightly higher than previously reported counts from cross sections of the wing nerve (455–465 axons, (Edwards et al., 1978 [DOI](#))). Of these afferents, we classified 364 as wing margin bristle axons, based on their ventral projections (Palka et al., 1979 [DOI](#)). Of the 126 non-bristle afferents, we identified 64 axon morphologies from published images of dye-fills (Table 1 [DOI](#); (Burt and Palka, 1982 [DOI](#); Ghysen, 1980 [DOI](#), 1978 [DOI](#); Kays et al., 2014 [DOI](#); Koh et al., 2014 [DOI](#); Lu et al., 2012 [DOI](#); Palka et al., 1986 [DOI](#), 1979 [DOI](#); Thistle et al., 2012 [DOI](#); Whitlock and Palka, 1995 [DOI](#))). Of the 62 remaining axons previously unidentified in the literature, we identified sparse GAL4 lines in the FlyLight collection (Jenett et al., 2012 [DOI](#)) that labeled axon morphologies that resembled the reconstructed axons from FANC, crossed these lines to a fluorescent reporter, and then imaged the wing and wing hinge to visualize expression (Figure 1S [DOI](#)). Using this strategy, we successfully identified 50 of the 62 previously unidentified morphologies.

We reconstructed postsynaptic partners of sensory neurons until at least 70% of the output synapses from each sensory neuron were attached to proofread neurons (Figure 2A [DOI](#)). Sensory axons make direct synapses onto motor neurons, other sensory neurons, VNC intrinsic neurons, and interneurons that ascend to the brain. To identify clusters of sensory axons with similar postsynaptic connectivity, we used a pairwise measure of cosine similarity, where a score of one indicates that the two neurons contact the same partners with the same proportion of synapses. We then ordered the neurons via agglomerative clustering, which revealed clusters of neurons with similar morphologies (Figure 2B [DOI](#)). The cosine similarity of axon pairs within each cluster was significantly higher than across clusters (Figure 2B [DOI](#), inset (permutation test; 10,000 permutations, observed difference = 0.34, $p < 0.05$)). Figure 2C–E [DOI](#) show the axon morphology of each cluster, organized by peripheral class. In the remainder of the paper, we focus on identifying the novel sensory neuron classes in Figure 2C [DOI](#).

Proximal CS axons are characterized by three branches: one short branch projects to the tectulum and two long branches project anteriorly to the brain and posteriorly to the haltere neuropil (Ghysen, 1980 [DOI](#)). There are ~36 proximal CS on the wing, and we found 38 axons in the EM dataset that followed this pattern. Previous dye fills of distal CS revealed axons that do not ascend to the brain, and instead send two processes to the posterior VNC (Ghysen, 1980 [DOI](#)). There are ~17 distal CS on the wing, and we found 15 axons that match this pattern. We also identified five ascending axons that resemble the small CS morphology although they are missing a posterior branch. Overall, our comprehensive reconstruction revealed many morphological subgroups with overlapping postsynaptic partners, suggesting a high degree of integration within wing sensorimotor circuits.

Table 1. ADMN Sensory axon identification and nomenclature

Sensory structure	Axon morphology identification	MANC connectome nomenclature (From Marin et al., 2024)
small campaniform sensilla (proximal)	Palka, 1979; Ghysen, 1980	<i>SApp04, SApp10, SApp11, SApp13, SApp14, SApp18, SApp19, SApp20, SApp21</i>
small campaniform sensilla (distal)	Palka, 1979; Ghysen, 1980	<i>SNpp04, SNpp08, SNpp11, SNpp33, SNpp36, SNpp06, SNpp26</i>
large campaniform sensilla	Burt and Palka, 1982; Palka, 1986	<i>SNpp30, SNpp32, SNpp31</i>
tegula campaniform sensilla	Lesser, 2025	<i>SNpp28, SNpp37, SNpp38</i>
tegula hair plate	Lesser, 2025	<i>SNxx26</i>
tegula chordotonal organ	Lesser, 2025	<i>SNpp07, SNpp10</i>
radius chordotonal organ	Lesser, 2025	<i>SNpp29, SNpp61, SNpp63, SNpp62</i>
thorax sensor	Lesser, 2025	<i>SNpp16</i>
thorax macrochete	Ghysen, 1980; Kays 2014	<i>SNta05, SNta10, SNta12, SNta13</i>
thorax microchete	Ghysen, 1978; 1980	<i>SNta01, SNta02, SNta09</i>
margin mechanosensors	Palka, 1979	<i>SNta04, SNta06, SNta07, SNta08, SNta11, SNta14, SNta18</i>
margin chemosensors	Lu 2012; Thistle 2012; Koh 2014	<i>SNch02, SNch03, SNch04, SNch12</i>
unknown	-	<i>SNpp05, SNpp09, SNpp13, SNpp27, SNxx28, SNtaxx, SNxxxx, SNxx24, SNxx25</i>

Table 2. Literature characterizing peripheral anatomy of wing mechanosensory neurons

Paper	Structures Identified
Hertweck, 1931	Chordotonal organ in the radius, thorax sensor
Fudalewicz-Niemczyk, 1963	Neuronal innervation in wings of ten dipteran species
Cole & Palka 1982	Differences in peripheral morphologies of wing campaniform sensilla domes; Homologies between wing campaniform sensilla and haltere campaniform sensilla
Hartenstein & Posakony, 1989	Comprehensive identification of wing and thorax bristles throughout development
Dinges et al., 2021	Comprehensive atlas of all campaniform sensilla in <i>Drosophila melanogaster</i> .

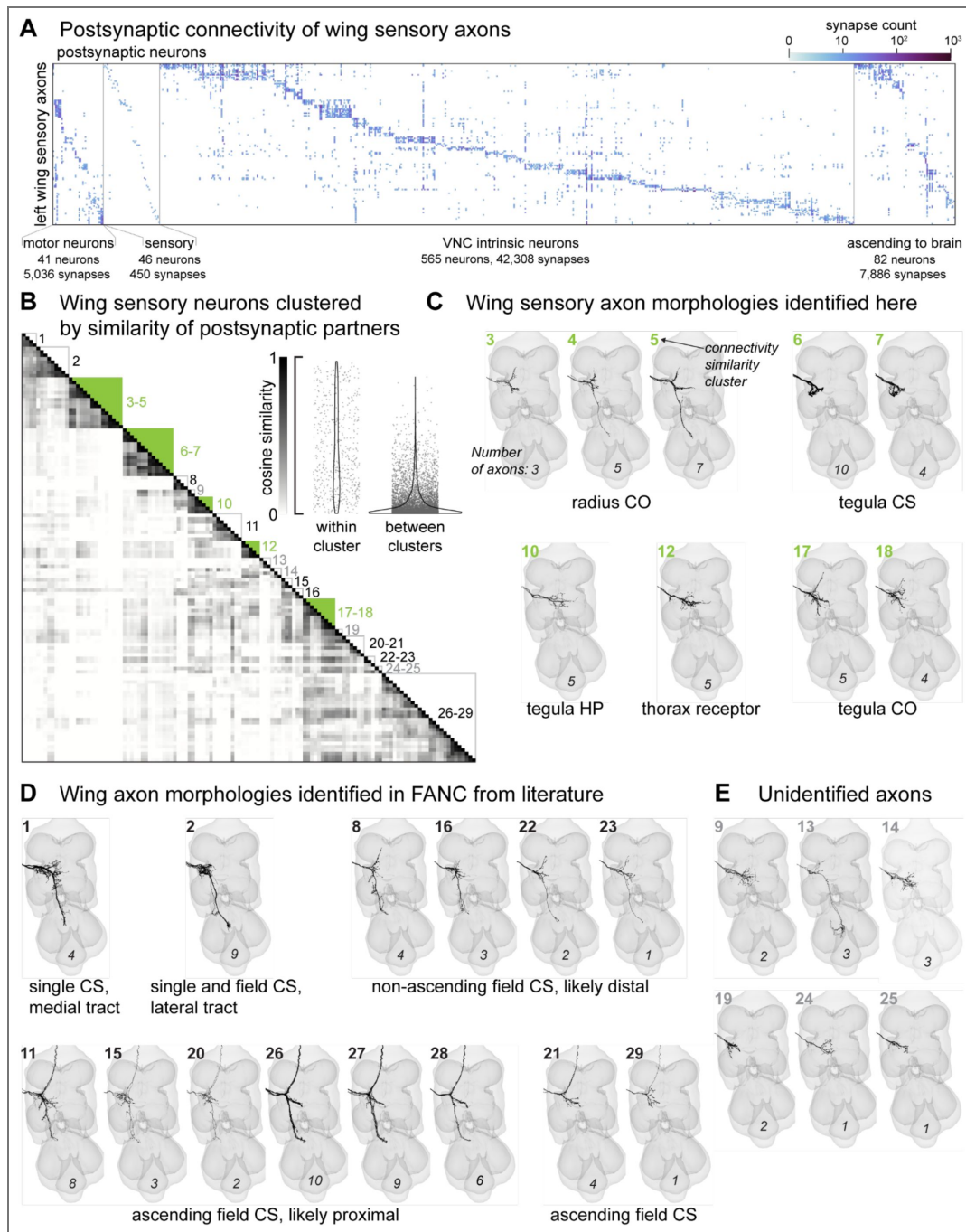


Figure 2. Postsynaptic connectivity and morphology of wing sensory axons.

(A) Connectivity matrix based on the left wing proprioceptors and postsynaptic neurons in the VNC. Only partners with at least five synapses from a single proprioceptor are shown. For visualization simplicity, we do not show: (1) a descending neuron that is postsynaptic to sensory neurons (0.1% of the proprioceptive outputs), (2) a single non-motor efferent neuron (0.1%); and unproofread or fragment neurons, (9.7%). Postsynaptic neurons are classified as either motor neurons, sensory neurons, VNC intrinsic neurons, or ascending neurons (axons project to the brain). Within each class, postsynaptic neurons were then sorted according to which wing proprioceptor they receive the most synapses from. The number of synapses is displayed on a log scale. (B) Cosine similarity matrix of the 126 left wing axons not from margin bristles. Axons are ordered by agglomerative clustering. Inset shows pairwise similarity scores for each pair of axons. Within cluster similarity is greater than between clusters (permutation test; 10,000 permutations, observed difference = 0.34, $p < 0.05$). Boxes in the matrix indicate clusters of axons with similar morphology, with the number next to each cluster indicating the morphology clusters in (C-E). Filled green boxes indicate morphologies identified in this study. See Table 1 and Methods for details on matching axon morphologies to prior literature.

Connectivity reveals a potential role for the tegula in flight control

Due to the need for rapid sensory feedback necessary for flight control, we were especially interested in identifying axons with monosynaptic connections onto wing motor neurons. We found that 34 (of 62) previously uncharacterized axons synapse directly onto wing steering motor neurons (Figure 3A–B). Of these, one group of axons synapses directly onto the well-characterized b1 motor neuron, which innervates the b1 muscle to help stabilize pitch during flight (Whitehead et al., 2022). The b1 motor neuron fires on nearly every wing stroke and input from wing afferents sets the phase of its activation (Fayyazuddin and Dickinson, 1999; Heide and Götz, 1996). Notably, the input to the b1 motor neuron from ipsilateral wing and haltere axons is clustered around the putative spike initiation zone (Figure 3B), as has previously been reported based on axonal spatial overlap (Chan and Dickinson, 1996). This synaptic organization may be a structural mechanism for facilitating rapid modulation of b1 activity based on sensory feedback. The wing sensory axons that synapse directly onto the b1 motor neuron have not been previously characterized (Figure 3C). They terminate shortly after entering the VNC and do not branch more extensively. We observed an unexpected ultrastructural feature in these axons: their terminals contain very densely packed mitochondria compared to other cells (Figure 3D). This feature is also present in interneurons with known electrical connections (Trimarchi and Murphey, 1997) to the b1 motor neuron (Figure 3E). We speculate on this ultrastructure further in the **Discussion**.

To identify the peripheral identity of these axons (Figure 3F), we found a driver line that labeled this population (Figure 3G; (Jenett et al., 2012)) and crossed it to a fluorescent reporter. Imaging the wing revealed that the population of short axons that directly synapse onto a subset of wing steering motor neurons originate from a field of CS on the tegula (Figure 3H). This finding suggests that the tegula may play a previously underappreciated role in flight control, particularly in regulating the tonically firing muscle b1 and a tonically firing muscle from another motor module, i2 (Figure 3A).

Hair plate (HP) on the tegula

A group of five axons branches dorsally and ventrally as they enter the nerve cord and cross the midline (Figure 4A). We identified a sparse driver line that labeled these neurons (Figure 4B) and found that their corresponding cell bodies were in the tegula. There is a row of short stubby hairs on the dorsal face of the tegula (Figure 4C), resembling the HPs found at leg joints, which are activated at extreme joint positions (Pratt et al., 2025; Pringle, 1938b; Trimarchi et al., 1999). The role of this tegula HP in wing sensation is unknown, although their peripheral morphology had been previously described (Fudalewicz-Niemczyk, 1963).

Chordotonal organ in the tegula

Two groups of axons with similar postsynaptic partners branch broadly throughout the tectulum without crossing the midline (Figure 5A). Using a sparse driver line (Figure 5B), we found that the cell bodies belong to an internal structure within the tegula (Figure 5C). We counted ~14 neurons in this structure, which separate into two bundles that attach to different points on the distal, anterior end of the tegula. Neurons in the chordotonal organ (CO) in the tegula are not labeled by *iav*-GAL4, unlike many other COs in the fly's body (Figure 5D; (Kwon et al., 2010)). They do, however, have actin-rich cap cells that are characteristic of other COs and not present in other mechanosensory neurons like CS or HPs (Figure 5E; (Field and Matheson, 1998)). A subset of tegula CO neurons are labeled by *NompC*-GAL4, suggesting that they do express mechanosensory channels other than *iav* (Figure 5F). Because we only identified nine axons, vs. 14 cell bodies, some tegula CO axons might have a different morphology.

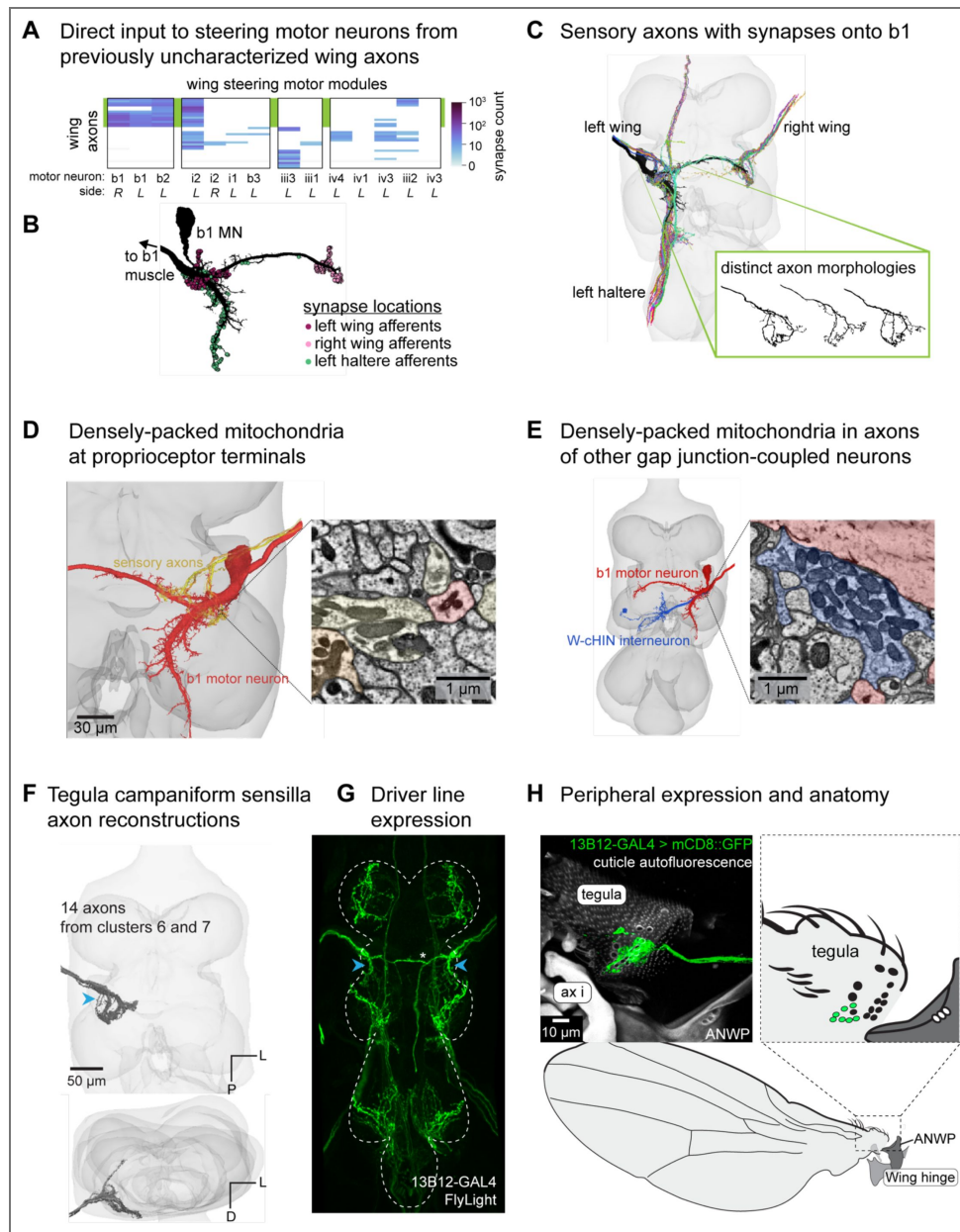


Figure 3. Campaniform sensilla (CS) on the tegula target the tonic wing steering motor neuron b1.

(A) Connectivity between previously uncharacterized wing sensory axons and wing steering motor neurons. Wing steering motor neurons (columns) are grouped by motor modules, which are groups of motor neurons that receive a high degree of synaptic input from shared presynaptic partners and are therefore likely to be co-activated (Lesser et al., 2024). The green box behind the plot highlights a group of axons with a shared morphology, discussed in the rest of the figure. (B) The left b1 motor neuron with circles showing predicted synapse locations from the FANC EM volume. (C) 3D reconstructions of the left b1 motor neuron (black) and all the sensory axons from which it receives direct synaptic input. Inset: three example individual axons from the left wing to demonstrate the variation in axon branching. (D) Ultrastructure of putative electrical synapses: these sensory axons feature densely packed mitochondria at terminals near the b1 motor neuron. (E) A similarly high density of mitochondria is also seen at axon terminals of a wing contralateral haltere interneuron (w-ChiN), which likely have electrical synapses onto b1 based on dye-fill experiments (Trimarchi and Murphey, 1997). (F) Axon branching pattern in VNC. Axons are from two morphological clusters (#6 and #7 from Figure 2). Below: rotated view of the VNC. (G) Maximum projection from FlyLight Z-Stack of images of the driver line 13B12-GAL4. Projection that is crossing the midline (indicated by a white asterisk) is from a different sensory neuron that enters through the Posterior Dorsal Mesothoracic Nerve and innervates a thorax bristle. (H) Expression in the periphery. Maximum projection from confocal z-stack showing sensory neurons that innervate the CS field on the tegula. The driver line also labels two tegula HP hairs, but their axon morphology is distinct (see Figure 4). Wing hinge abbreviations: anterior nodal wing process (ANWP), first axillary (ax i).

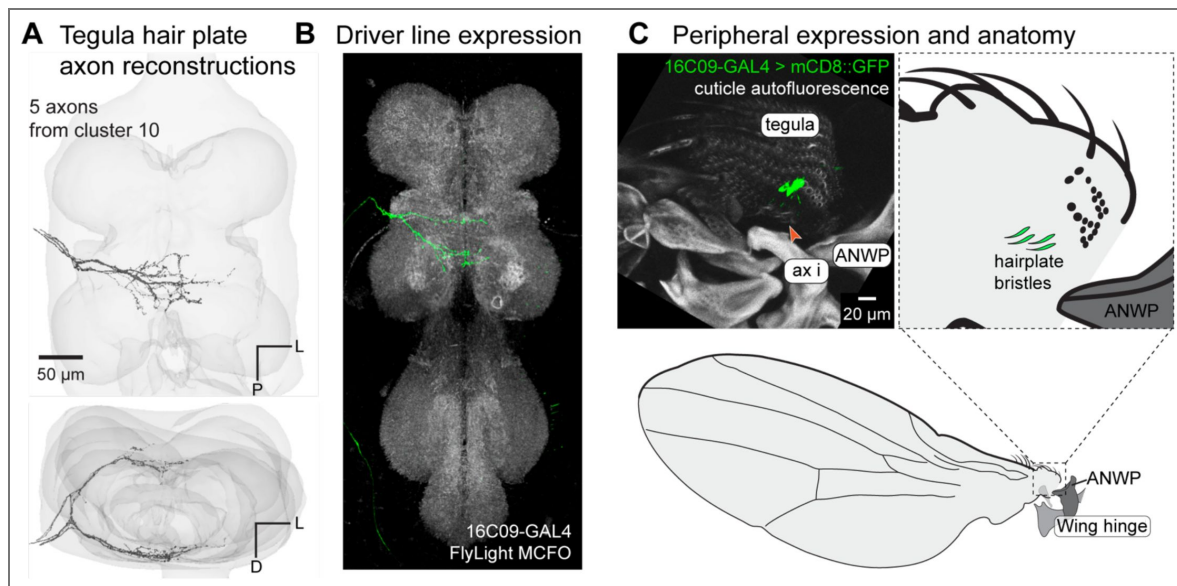


Figure 4. Tegula hair plate.

(A) 3D reconstructed axons. Above: population of axons with similar morphology (black) and VNC volume (gray). Below: rotated view to show how the axons split to scoop around the dorsal and ventral edges of the wing neuropil. (B) Axon branching pattern in VNC. Axons are from morphological cluster #10 in Figure 2. Maximum projection from the FlyLight MCFO collection of the driver line 16C09-GAL4. (C) Expression in the periphery. Maximum projection from confocal Z-Stack showing sensory neurons that innervate the hairs of the tegula hair plate. Red arrow indicates an external hair plate hair. Wing hinge abbreviations: anterior nodal wing process (ANWP), first axillary (ax i).

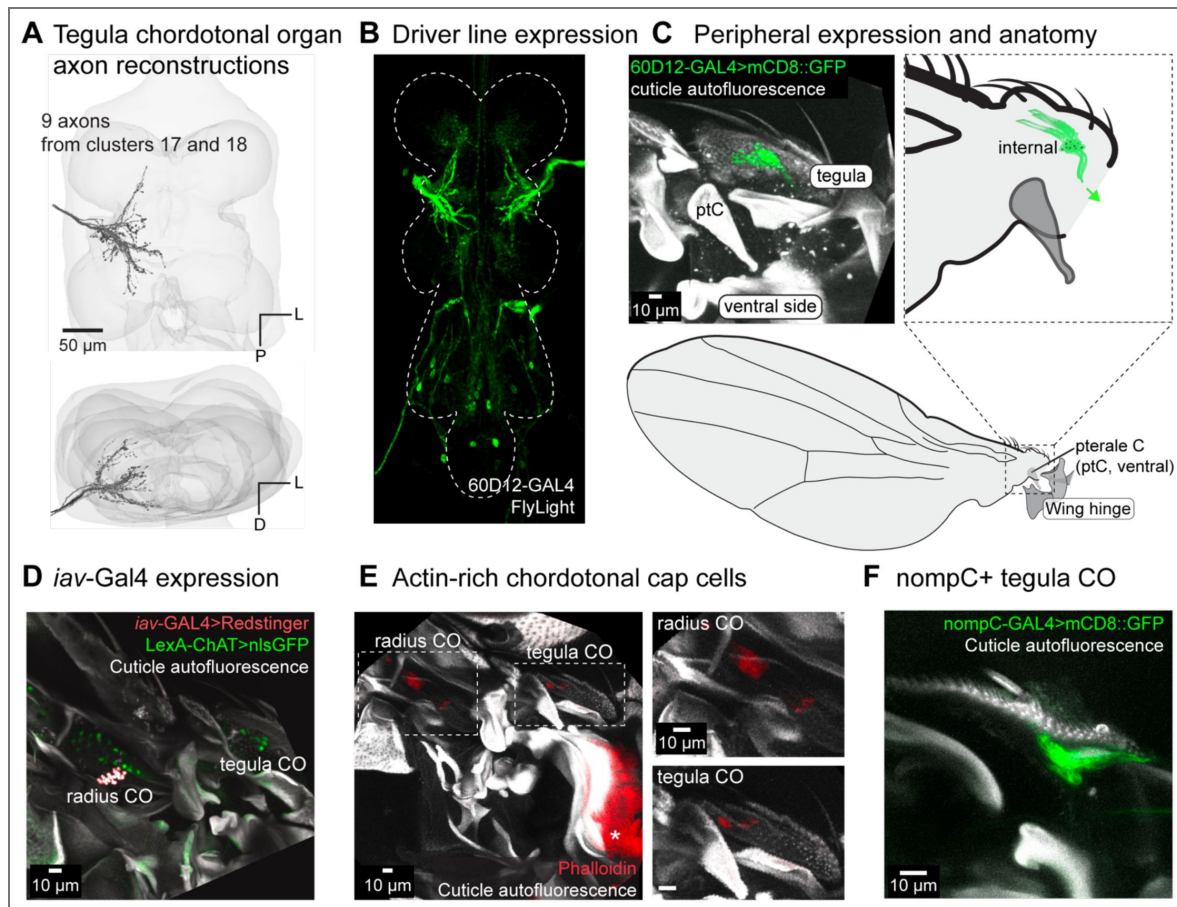


Figure 5. Tegula chordotonal organ.

(A) 3D reconstructed axons. Axons are from two morphological clusters (#17 and #18 in Figure 2). (B) Axon branching pattern in VNC. Maximum projection from FlyLight Z-Stack of images of the driver line 60D12-GAL4. (C) Expression in the periphery. Maximum projection from confocal Z-Stack showing sensory neurons that innervate the chordotonal organ in the tegula. There are two clusters of neurons, which are differentiated by their separate attachment points within the tegula. 60D12-GAL4 labels neurons from both clusters. (D) Maximum z-projection of the proximal wing co-labeling *iav*-GAL4 with ChAT-LexA. ChAT-LexA labels nearly all sensory neurons (green, nuclear stain) and *iav*-GAL4 labels the radius CO but not the tegula CO (red, nuclear stain). (E) Phalloidin labels the actin-rich cap cells that are part of chordotonal organs. Asterisk indicates muscle that is also labeled by phalloidin. (F) *nompC*-GAL4 labels all sensory neurons in the tegula, including the chordotonal organ.

Chordotonal organ in the radius

Axons with three distinct morphologies share a characteristic branch that passes laterally along the bottom of the wing neuropil (Figure 6A). One group of axons extends a long process into the haltere neuropil, and another crosses the midline. By imaging sparse driver lines, we found that these axons come from neurons that make up a CO in the radius (Figure 6B-C). These neurons are distinguishable from the CS neurons in the radius because they do not have a dendrite that reaches toward the surface of the vein to innervate a dome. Instead, the cell bodies sit on the posterior side of the radius and their dendrites and cap cells insert on the anterior side of the radius. These neurons all attach to the same point on the wing vein (Figure 6D), so they are likely subject to the same mechanical forces, allowing the CO to send parallel information to multiple regions of the VNC.

Thorax sensor near the wing hinge

A population of five axons extends single processes through the dorsal tectulum, and two of the axons cross the midline (Figure 7A). These axons originate from a cluster of five neurons in the thorax beneath the wing hinge near the parascutal shelf, just medial to the anterior nodal wing process (Figure 7B-C). Other than the three CS on the anterior nodal wing process, these are the only cells near the wing hinge labeled by the *ChAT-GAL4* driver line, which targets nearly all peripheral sensory neurons (Figure 7C) (Yasuyama and Salvaterra, 1999). As with all the anatomically-defined populations of axons, the function of these novel wing hinge sensory neurons will require physiological measurements, but based on their location they may signal wing opening and closing. We saw no evidence for sensory neurons innervating pterale C (Figure 7D), a wing hinge sclerite that was previously thought to contain sensory receptors (Miyano and Ewing, 1984), although axons from the radius travel directly beneath pterale C.

Adjacent campaniform sensilla (CS) axons are morphologically distinct

Previous work uncovered morphological diversity across CS axons, where subsets of CS axons follow different tracts and some ascend to the brain while others do not (Palka et al., 1979). It was unclear, however, whether there is morphological diversity of axons that originate from the same CS field. A field of CS are defined as having the same size, circularity, orientation, and general location on the wing (Cole and Palka, 1982). Single CS can be either rapidly or slowly adapting (Dickinson and Palka, 1987), but whether all CS within a field have the same firing properties remains unknown. It is also not known whether all CS within a field connect to the same postsynaptic partners. Answering this question could provide insight into the function of spatially clustered CS, for example whether they underlie a population code or transmit signals in parallel to distinct downstream circuits.

To determine the morphological similarity of axons that innervate CS in the same field, we identified GAL4 driver lines that sparsely expressed in the wing nerve (less than five ADMN axons) and imaged their expression in the wing. We found many lines that label subsets of CS across multiple fields (Table 3) and fortuitously found three driver lines that label one to two separate CS in the ventral radius C field (v.Rad.C, Figure 8A). For these three lines, we compared VNC expression of axons from the ADMN using the FlyLight MCFO collection (Figure 8B-D; (Meissner et al., 2023)). In a driver line that expresses in two v.Rad.C neurons (CS 2 and 4), the VNC contains two distinct axon morphologies originating from the ADMN nerve (Figure 8D, row 1). Both axons possess a process that ascends to the brain, but one also projects down to the haltere neuropil. In a second driver line that also labels the second CS in v.Rad.C, we observed the same axon morphology that ascends to the brain but does not reach the haltere neuropil (Figure 8D, row 2). A third driver line, that expresses in the third CS in v.Rad.C, contains a non-ascending wing axon with two posterior projections (Figure 8D, row 3). These results show that neurons that innervate adjacent campaniform sensilla within the same field can have different axon morphologies. Each of the three v.Rad.C axons falls into a cluster of morphologically similar axons

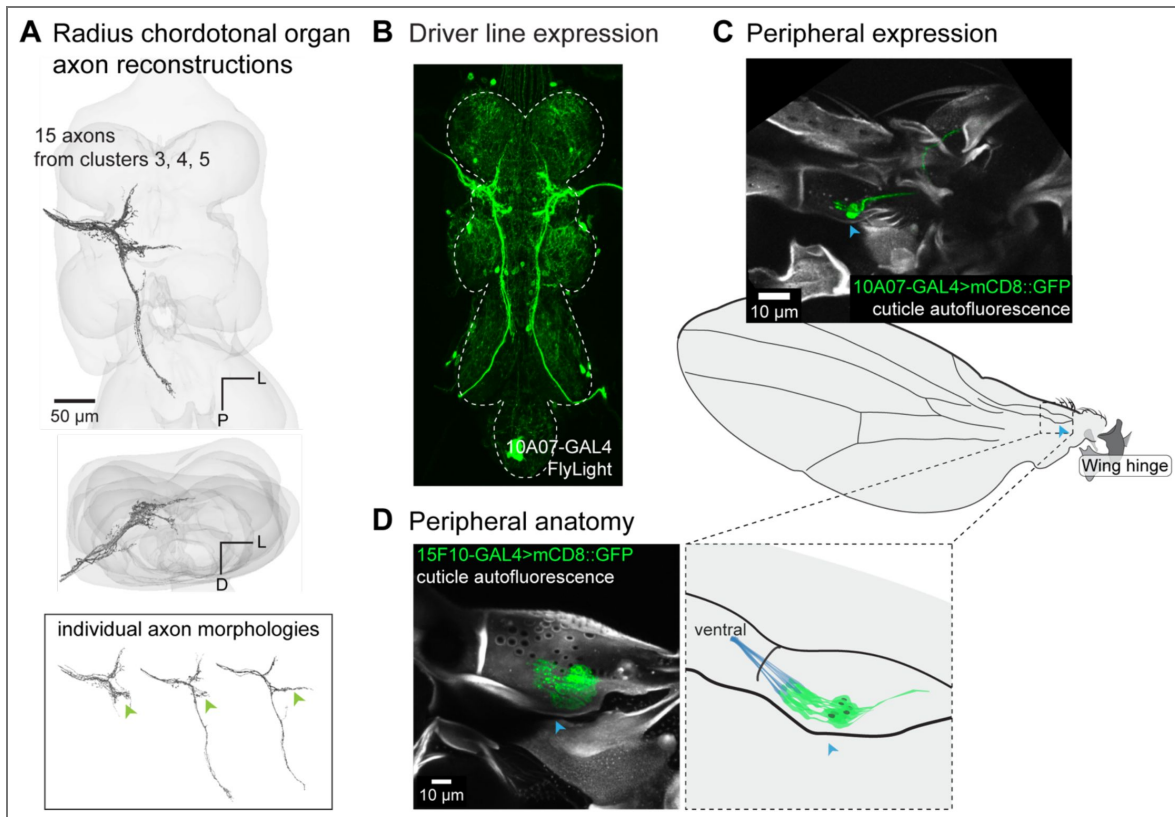


Figure 6. Radius chordotonal organ.

(A) 3D reconstructed axons. Axons are from the morphological clusters #3, #4, and #5 in Figure 2. Green arrow indicates the characteristic lateral projection found in each neuron. (B) A sparse driver line, 10A07-GAL4, labels a subset of neurons that make up the radius chordotonal organ. For other driver lines that label radius chordotonal neurons, see Table 3. (C) Peripheral expression of 10A07-GAL4 > UAS-mCD8::GFP. (D) Peripheral anatomy of the radius chordotonal organ, which is better shown by a broad driver line, 15F10-GAL4 > UAS-mCD8::GFP. The radius chordotonal organ attaches to the ventral inner wall of the radius by cap cells (blue). A blue arrow is shown across the confocal images and cartoons to orient to the “pocket” in the radius near the CO cell bodies.

Peripheral structure	# of neurons	total	Radial CO	Tegula CO	ANWP CS	tegula CS field	d.Rad.A	d.Rad.B	d.Rad.C	v.Rad.A	v.Rad.B	v.Rad.C	d.Rad.D	d.Rad.E	dS-1 & 2	d.HCV	GSR	TSM-1 & 2	ACV	L3-1	L3-2	L3-3	v.HCV	L3-V	thorax receptor	tegula hair plate	tegula hairs
			~24	~14	3	18	4	7	18	4	3	5	4	8	2	1	1	2	1	1	1	1	1	1	5	5	6
10A07-Gal4	4	4																									
10F07-Gal4	8	7					1																				
10G03-Gal4	9							2		2														5			
12C07-Gal4	19						2	10		1	3		2									1					
13B12-Gal4	10				8																				2		
15F10-Gal4	24	24																									
16C09-Gal4	4																								4		
21A01-Gal4	5		5																								
21C09-Gal4	7							2	4		1																
24C04-Gal4	8	6									2																
26B11-Gal4	2																2										
26D04-Gal4	12							5	2		3		2														
26F04-Gal4	36					4	7	17		2		4	2														
35B08-Gal4	3	3																									
36C09-Gal4	4	4																									
37D11-Gal4	8			1																				5		2	
38H01-Gal4	2										1											1					
39F05-Gal4	1	1																									
42G08-Gal4	2				2																						
44H11-Gal4	9	8	1																								
48H11-Gal4	4	4																									
49F11-Gal4	7	7																									
54H12-Gal4	9			2	2									2										3			
57F03-Gal4	5		5																								
60B12-Gal4	4											2									1	1					
60D12-Gal4	10		10																								
60G04-Gal4	5		5																								
64C04-Gal4	6	4				1		1																			
70G12-Gal4	2																								2		
72C01-Gal4	6	6																									
73F02-Gal4	2				2																						
75B09-Gal4	21		4	2	13		2																				
76E12-Gal4	8	8																									
79G12-Gal4	1										1																
83B04-Gal4	5																							5			

Table 3. Driver lines labeling wing sensory neurons

Numbers in the table indicate how many neurons are labeled, e.g., 4 of 24 radius CO neurons for the first driver line, 10A07-GAL4.

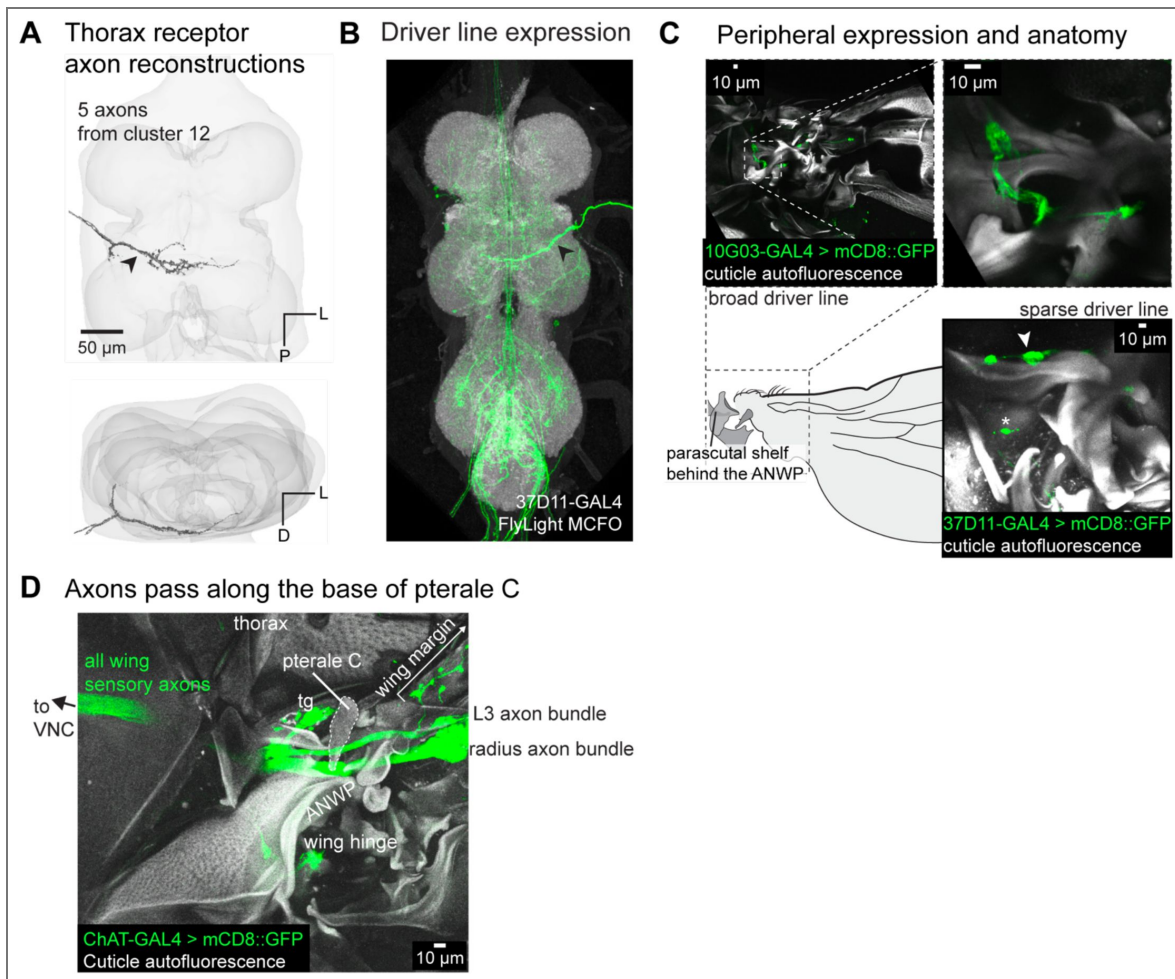


Figure 7. Sensory axons near the wing hinge.

(A) 3D reconstructed axons. Axons belong to the morphological cluster #10 from Figure 2. (B) Axon branching pattern in VNC. Maximum projection from the FlyLight MCFO collection of the driver line 37D11-GAL4. (C) Expression in the periphery. Top: maximum projection from confocal Z-Stack of a broader driver line, 10G03-GAL4, to show the morphology of the sensory neurons at the base of the parascutal shelf. Below: maximum projection from confocal Z-Stack of the sparse driver line 37D11-GAL4 > UAS-mCD8::GFP showing neurons labeled at the base of the parascutal shelf. The asterisk marks an innervated bristle on the thorax. (D) Pterale C is not an innervated sclerite. Pterale C was previously predicted to be innervated based on experiments in which an electrode placed at the base of pterale C recorded signals in response to wing vibration (Miyan and Ewing, 1984). We found no neurons innervating pterale C, but we did observe that the axon bundle from the radius passes directly under pterale C, which could explain previously published results.

that connect to similar postsynaptic neurons (Figure 8E [↗](#), clusters 2, 15, and 21, respectively, from Figure 2 [↗](#)). The additional axons in each cluster likely originate from CS in other fields, on different parts of the wing. Overall, our results suggest that adjacent CS neurons in the same field connect to different target neurons in the VNC, and that the CS from different fields can connect to common postsynaptic targets.

Discussion

In this project, we reconstructed each sensory axon in the left ADMN wing nerve, as well as their postsynaptic partners, from an EM dataset of the *Drosophila* VNC (Azevedo et al., 2024 [↗](#); Phelps et al., 2021 [↗](#)). We identified their postsynaptic partners in the EM volume and matched previously unidentified axon morphologies to their sensory structures on the wing by imaging the expression of genetic driver lines. To make this information accessible to the community, we provide a library of confocal Z-stacks and an annotation table linked to the FANC connectome dataset (see **Methods**). We also include a reference (Table 1 [↗](#)) to match peripheral sensory structures identified here to neuron nomenclature established in the male VNC connectome (MANC), in which wing sensory axons are proofread but few are annotated by peripheral identity (Marin et al., 2024 [↗](#)). The relative ease with which we were able to match sensory neurons across the two datasets based on morphology suggests that there were no obvious differences at level of sexual dimorphism. A closer examination across more than two datasets is needed to rigorously support this observation, though.

Integrating connectomics with previous research

Sensory neurons in the *Drosophila* wing have been a useful model system for investigating the relative contributions of intrinsic regulation vs. extrinsic signaling in the determination of axon morphology (Ghysen, 1980 [↗](#); Palka et al., 1979 [↗](#)). Past work identified the axonal morphologies of wing CS and differences in the genetic programs guiding axon development for CS in fields and single CS (Ghysen, 1980 [↗](#); Palka et al., 1979 [↗](#)). Here, we build on this knowledge using cell-type specific genetic tools and connectomics to create a more complete map of the wing sensory apparatus and central sensory circuits. By reconstructing axons from an electron microscopy volume, we found novel wing sensory axon morphologies, many of which originated from internal sensory structures that were inaccessible with previous dye-fill techniques, such as the previously uncharacterized CO axons.

Most wing CS are organized into fields, in which domes of roughly the same size and shape cluster together. By finding sparse driver lines that label only one or two CS in the ventral radius C field (v.Rad.C), we found that CS in the same field can have distinct axon morphologies. This finding is consistent with our unsuccessful attempts to build split-GAL4 driver lines that specifically label CS from a single field by intersecting lines which label the same axon morphology. This organization may offer a functional advantage by sending similar signals to multiple regions of the CNS in parallel; viewed from a different perspective, postsynaptic neurons may rapidly integrate information across different CS fields. One possibility is that CS axon morphology is more closely linked to neuronal intrinsic properties (e.g., slowly vs. rapidly adapting). We hypothesize that the morphological clusters we identified (Figure 8D-E [↗](#)) are similarly tuned CS neurons distributed throughout different fields across the wing. A concurrent project that focused on the central and peripheral organization of haltere mechanosensory neurons found that common central axon morphologies map across multiple CS fields on the haltere (Dhawan et al., 2025 [↗](#)), consistent with our prediction for the wing.

One advantage of the dye-fill technique is that it is sometimes feasible to repeatedly label the same exact neuron across individuals. By comparing axon morphologies across individuals, researchers have found differences in small branches that extend from the primary neurite of the neuron (Kays et al., 2014 [↗](#); Palka et al., 1986 [↗](#)). As more connectomic datasets become available, we will be able to ascertain if this morphological variation is also reflected by variation in downstream

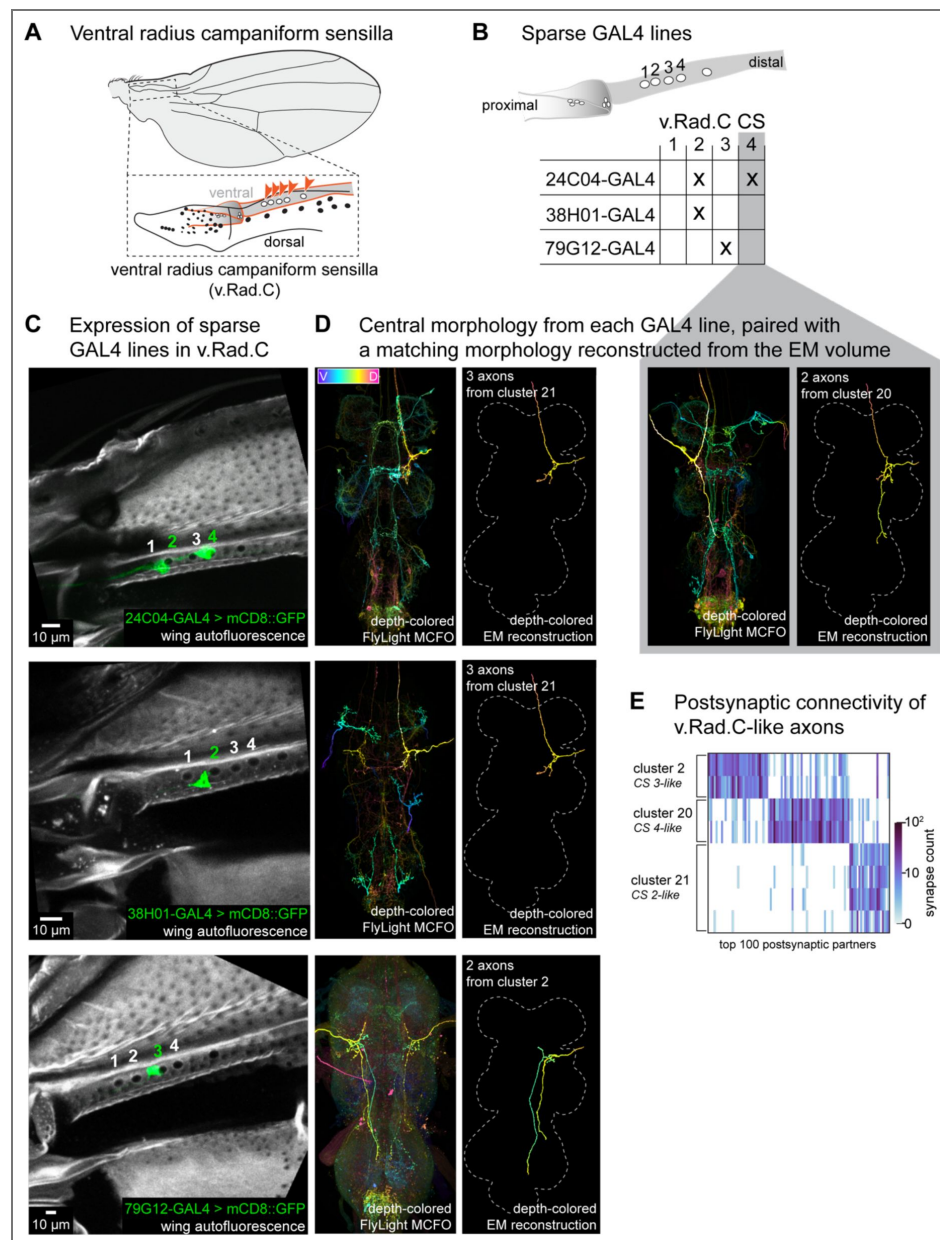


Figure 8. Campaniform sensilla in the same field have unique axons.

(A) The ventral radius C (v.Rad.C) field of campaniform sensilla is on the ventral side of the more distal part of the radius. The field has four to five domes, the fifth dome is proposed to be its own individual dome as it is farther apart from the other four and its orientation is slightly different (Dinges et al., 2021). (B) Summary of the CS within v.Rad.C that are labeled by sparse GAL4 lines shown in C. (C) Peripheral expression in specific campaniform sensilla from sparse driver lines (each row). Maximum projection from confocal z-stack showing expression in the periphery from each sparse driver line. CS in v.Rad.C are labeled one through four, as in (B) to show which CS is innervated in each image. (D) Pairs of images showing (left) A depth-colored single channel MCFO Z-Stack from the FlyLight collection (Meissner et al., 2023), with the wing axon highlighted in the image. Contrast of z-sections was optimized to emphasize visual clarity of wing axons, see **Methods** for details. (right) The reconstructed axon from EM that best matches the morphology, depth-colored and aligned to the same template as the FlyLight images. (E) Postsynaptic connectivity of axons with morphologies that match those found for v.Rad.C. Postsynaptic connectivity is more similar for axons with similar morphologies than from the same CS field.

connectivity. Understanding relationships between morphology and connectivity across individuals, and even across species, will provide a framework for deciphering the developmental logic that governs the formation of sensorimotor circuits.

Insights into motor control based on connectivity and structural anatomy

Direct sensory input to wing motor neurons

Many wing sensory neurons, such as those originating from the tegula and near the wing hinge, synapse directly onto wing motor neurons, providing a mechanism for rapid feedback-driven motor control. Four steering wing motor neurons receive over 10% of their synaptic input from sensory neurons, including haltere input (Lesser et al., 2024 [DOI](#)). Each of these four motor neurons also fires tonically during flight, whereas other steering MNs burst during maneuvers such as turning (Lindsay et al., 2017). Integrating direct sensory input provides a mechanism for low latency motor control, which is important given the short timescales required for Dipteran flight (Dickerson, 2020 [DOI](#)). We found that the CS on the tegula provide the most direct feedback onto wing steering motor neurons (Figure 3 [DOI](#)). Future work is needed to understand what these sensory neurons encode, but based on their direct connections and thick axons, we speculate that they set the firing phase of tonically active muscles such as b1 (Fayyazuddin and Dickinson, 1999 [DOI](#); Heide, 1983 [DOI](#)).

The tegula as a major sensory structure

Though the tegula has been studied for its role in locust flight control, it has largely been neglected in Dipteran literature. In both locusts and Diptera, the tegula features CS, a CO, and mechanosensory hairs (Fudalewicz-Niemczyk, Władysława, 1963 [DOI](#); Wolf, 1993 [DOI](#)). Feedback from the locust tegula resets the phase of wing elevation– the forewing tegulae are only necessary to maintain the flight rhythm if the hindwing tegulae are compromised (Büschges et al., 1992 [DOI](#)). Unlike in locusts, Dipteran flight muscles are asynchronous, where wing elevation and depression during each wing stroke are biomechanically rather than neurally controlled (Deora et al., 2015 [DOI](#)). Despite the different feedback demands between asynchronous and synchronous flight, all flying insects share an evolutionary history (Gau et al., 2023 [DOI](#)) and wings experience similar forces, which are optimally sensed from particular locations (Weber et al., 2021 [DOI](#)). Examining how the tegula contributes to flight control across taxa offers opportunities to better understand the evolutionary pressures shaping mechanosensory feedback in flying insects.

Mechanosensation at the wing hinge

In locusts, feedback from stretch receptors embedded in the wing hinge can directly modify wingbeat frequency (Gettrup, 1962 [DOI](#); Wilson and Gettrup, 1963 [DOI](#)). In *Diptera*, however, there is scant literature on a sensory organ embedded near the wing hinge (Hertweck, 1931 [DOI](#)). The only putative sensory structure at the wing hinge is the sclerite pterale C (Miyan and Ewing, 1984 [DOI](#)). This hypothesis was based on spikes recorded from a sharp electrode placed at the base of pterale C in response to wing vibration. We found no cells labeled by a pan-sensory neuron driver (*ChAT-GAL4*) at the base of, or innervating, sclerite pterale C. We did, however, observe that the entire nerve of sensory axons from the radius passes through the base of pterale C (Figure 7D [DOI](#)), and thus speculate that action potentials traveling along this nerve are likely what was being recorded in that prior study. We also observed a cluster of previously unreported cells labeled by *ChAT-GAL4* near the parascutal shelf, which was also labeled by several sparse driver lines (see Table 3 [DOI](#), column “thorax receptor”), and may be the same structure described previously (Hertweck, 1931 [DOI](#)).

A potential metabolic specialization for flight circuitry

In addition to morphology and connectivity, the EM volumes of the VNC reveal the ultrastructure of neurons and synapses. While reconstructing neurons in FANC, we noticed an unusual density of mitochondria in the axon terminals of specific wing sensory and premotor neurons (Figure 3D-E). We did not notice equivalent specializations in prior projects that reconstructed and analyzed leg proprioceptors (Lee et al., 2025) and premotor neurons (Lesser et al., 2024). Notably, some of the terminals with dense mitochondria were at sites of known gap-junction coupling (Trimarchi and Murphey, 1997). In the fly VNC, electrical synapses are often accompanied by chemical synapses, which may cooperate to ensure low latency signal transmission (Fayyazuddin and Dickinson, 1996). In the adult fly brain, however, sites of gap junction coupling, such as the lobula plate tangential cells (Ammer et al., 2022), do not exhibit particularly high concentrations of mitochondria (Sager et al., 2024). Therefore, we speculate that the density of mitochondria we observe may be a specialization to support the energy-demanding low-latency feedback necessary for controlling flight. More work is needed to understand the significance and function of high mitochondrial density in wing sensorimotor circuits.

Remaining gaps

Although it is by far the most comprehensive to date, our atlas of wing sensory neurons is not complete. There were six axon morphologies (12 total axons) from the connectome that we could not reliably map to peripheral structures, as well as several peripheral structures whose axon morphologies we could not identify, such as the three CS on the anterior nodal wing process and the two large CS on the tegula. Further, some of the uncharacterized axon morphologies likely belong to the tegula and radius COs, both of which had more cell bodies than axons. These gaps highlight the need for complementary approaches, such as combining small-scale experimental approaches with large-scale comprehensive datasets, to fully characterize the wing's sensory landscape.

Methods

Reagent or resource	Source	Identifier
Antibodies		
Alexa Fluor Phalloidin 647	ThermoFisher A22287	n/a
Deposited data		
FANC connectome	Phelps et al., 2021; Azevedo et al. 2024	http://fanc.community
FlyLight	Meissner et al. 2024	https://www.janelia.org/project-team/flylight
Experimental models: <i>Drosophila</i> stocks		
Stock Name	Stock Source	Genotype
10A07-GAL4	RRID:BDSC_48435	w[1118]; P{y[+t7.7] w[+mC]=GMR10A07-GAL4}attP2
10F07-GAL4	RRID:BDSC_48266	w[1118]; P{y[+t7.7] w[+mC]=GMR10F07-GAL4}attP2
10G03-GAL4	RRID:BDSC_48271	w[1118]; P{y[+t7.7] w[+mC]=GMR10G03-GAL4}attP2
12C07-GAL4	RRID:BDSC_48496	w[1118]; P{y[+t7.7] w[+mC]=GMR12C07-GAL4}attP2
13B12-GAL4	RRID:BDSC_45796	w[1118]; P{y[+t7.7] w[+mC]=GMR13B12-GAL4}attP2
15F10-GAL4	RRID:BDSC_49266	w[1118]; P{y[+t7.7] w[+mC]=GMR15F10-GAL4}attP2
16C09-GAL4	RRID:BDSC_48720	w[1118]; P{y[+t7.7] w[+mC]=GMR16C09-GAL4}attP2
21A01-GAL4	RRID:BDSC_49853	w[1118]; P{y[+t7.7] w[+mC]=GMR21A01-GAL4}attP2
21C09-GAL4	RRID:BDSC_48936	w[1118]; P{y[+t7.7] w[+mC]=GMR21C09-GAL4}attP2
24C04-GAL4	RRID:BDSC_49072	w[1118]; P{y[+t7.7] w[+mC]=GMR24C04-GAL4}attP2
26B11-GAL4	RRID:BDSC_49164	w[1118]; P{y[+t7.7] w[+mC]=GMR26B11-GAL4}attP2
26D04-GAL4	RRID:BDSC_49175	w[1118]; P{y[+t7.7] w[+mC]=GMR26D04-GAL4}attP2
26F04-GAL4	RRID:BDSC_49191	w[1118]; P{y[+t7.7] w[+mC]=GMR26F04-GAL4}attP2
35B08-GAL4	RRID:BDSC_49818	w[1118]; P{y[+t7.7] w[+mC]=GMR35B08-GAL4}attP2
36C09-GAL4	RRID:BDSC_49933	w[1118]; P{y[+t7.7] w[+mC]=GMR36C09-GAL4}attP2
37D11-GAL4	RRID:BDSC_49536	w[1118]; P{y[+t7.7] w[+mC]=GMR37D11-GAL4}attP2
38H01-GAL4	RRID:BDSC_50025	w[1118]; P{y[+t7.7] w[+mC]=GMR38H01-GAL4}attP2
39F05-GAL4	RRID:BDSC_50056	w[1118]; P{y[+t7.7] w[+mC]=GMR39F05-GAL4}attP2
42G08-GAL4	RRID:BDSC_50166	w[1118]; P{y[+t7.7] w[+mC]=GMR42G08-GAL4}attP2
44G12-GAL4	RRID:BDSC_47933	w[1118]; P{y[+t7.7] w[+mC]=GMR44G12-GAL4}attP2
44H11-GAL4	RRID:BDSC_41268	w[1118]; P{y[+t7.7] w[+mC]=GMR44H11-GAL4}attP2
45D07-GAL4	RRID:BDSC_49562	w[1118]; P{y[+t7.7] w[+mC]=GMR45D07-GAL4}attP2
48H11-GAL4	RRID:BDSC_50396	w[1118]; P{y[+t7.7] w[+mC]=GMR48H11-GAL4}attP2
49F11-GAL4	RRID:BDSC_38701	w[1118]; P{GMR49F11-GAL4}attP2
54H12-GAL4	RRID:BDSC_48205	w[1118]; P{y[+t7.7] w[+mC]=GMR54H12-GAL4}attP2/TM3, Sb[1]
57F03-GAL4	RRID:BDSC_46386	w[1118]; P{y[+t7.7] w[+mC]=GMR57F03-GAL4}attP2
60B12-GAL4	RRID:BDSC_39239	w[1118]; P{y[+t7.7] w[+mC]=GMR60B12-GAL4}attP2
60D12-GAL4	RRID:BDSC_39249	w[1118]; P{y[+t7.7] w[+mC]=GMR60D12-GAL4}attP2
60G04-GAL4	RRID:BDSC_39258	w[1118]; P{y[+t7.7] w[+mC]=GMR60G04-GAL4}attP2
64C04-GAL4	RRID:BDSC_39296	w[1118]; P{y[+t7.7] w[+mC]=GMR64C04-GAL4}attP2
70G12-GAL4	RRID:BDSC_39552	w[1118]; P{y[+t7.7] w[+mC]=GMR70G12-GAL4}attP2
72C01-GAL4	RRID:BDSC_47729	w[1118]; P{y[+t7.7] w[+mC]=GMR72C01-GAL4}attP2
73F02-GAL4	RRID:BDSC_39824	w[1118]; P{y[+t7.7] w[+mC]=GMR73F02-GAL4}attP2
75B09-GAL4	RRID:BDSC_39883	w[1118]; P{y[+t7.7] w[+mC]=GMR75B09-GAL4}attP2
76E12-GAL4	RRID:BDSC_47753	w[1118]; P{y[+t7.7] w[+mC]=GMR76E12-GAL4}attP2
79G12-GAL4	RRID:BDSC_40051	w[1118]; P{y[+t7.7] w[+mC]=GMR79G12-GAL4}attP2
83B04-GAL4	RRID:BDSC_41309	w[1118]; P{y[+t7.7] w[+mC]=GMR83B04-GAL4}attP2
nompC-GAL4	RRID:BDSC_36361	y[1] w[*]; PBac{y[+mDint2] w[+mC]=nompC-GAL4.P}VK00014; Df(3L)Ly, sens[Ly-1]/TM6C, Sb[1] Tb[1]
UAS-mCD8::GFP	Gift from Rubin Lab, Janelia	P{pJFRC7-020XUAS-IVSmCD8::GFP}attP2

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, John C. Tuthill (tuthill@uw.edu [↗](#)).

Materials availability

The genetic driver lines used in this study are listed in [Table 3](#) [↗](#) and are available from the Bloomington *Drosophila* Stock center.

EM images & neuron reconstruction

The 3D reconstructed axons are from the FANC dataset (Phelps et al., 2021 [↗](#)), for details on segmentation, see [Azevedo et al., 2024](#) [↗](#). Only the left wing afferents were analyzed due to damage to the right side ADMN (Azevedo et al., Extended Data Figure 4). Following automatic segmentation, neurons were proofread to include primary neurites and as many branches as could confidently be reattached. Neurons were annotated using CAVE (Dorkenwald et al., 2023 [↗](#)). Depth-colored reconstructions were created using [braincircuits.io](#) [↗](#) (Azevedo et al., 2024 [↗](#)).

Reconstructed axon morphology clusters

To group axons by similar connectivity, we computed the cosine similarity of synaptic weights onto postsynaptic partners. We included fragments (9.7% of total output synapses) and used a three synapse threshold for connections. Cosine similarity and agglomerative clustering were computed with the python library Scikit-learn (cosine_similarity, AgglomerativeClustering, and dendrogram packages) (Pedregosa et al., 2011 [↗](#)). Permutation test to compare within- and between-cluster similarity was computed with the python library SciPy (Virtanen et al., 2020 [↗](#)). Information on synapse location predictions and error can be found in [Azevedo et al., 2024](#) [↗](#).

Animals

We used *Drosophila melanogaster* raised on standard cornmeal and molasses medium at 25 °C in a 14:10 hour light:dark cycle. We used female flies 2-7 days post-eclosion for imaging. The genetic driver lines screened are listed in [Table 3](#) [↗](#). Originally they were chosen by manually looking through the FlyLight database for driver lines with sparse expression in the ADMN. Later, the [braincircuits.io](#) [↗](#) “genetic lines matching” tool was used to screen driver lines by inputting segment IDs for particular wing afferent morphologies, with the particular lines selected based on sparse ADMN expression.

The fly food recipe used was based on the Bloomington standard Cornmeal, Molasses and Yeast Medium recipe, which can be found at

<https://bdsc.indiana.edu/information/recipes/molassesfood.html> [↗](#). Our recipe had only slightly different antifungal ingredients and included tegosept, propionic acid, and phosphoric acid.

Sample preparation

Wing images

To remove wings, flies were briefly anesthetized using CO₂ before using forceps to delicately cut around the wing hinge and remove the wing with the sclerites that make up the wing hinge intact. One wing was collected from 4-6 females to ensure that expression was consistent across individuals. Wings were then fixed in 4% formaldehyde (PFA) PBS solution for 20-60 minutes. Next, wings were rinsed in PBS with 0.2% Triton X-100 (PBT) four times over the course of 75 minutes. For most samples, native fluorescence was imaged, so the wings were then mounted onto slides in Vectashield without DAPI.

For preparations requiring immunohistochemistry to label cap cells of chordotonal organs, after rinsing, wings were incubated in 1:50 Alexa Fluor 635-nm Phalloidin (ThermoFisher A34054) in a PBS solution with the following reagents to improve tissue penetrance: 1% triton X-100, 0.5% DMSO, 0.05 mg/ml Escin (Sigma-Aldrich, E1378) and 3% normal goat serum. Wings were then

incubated for ten days at 4 °C overnight with gentle nutating at room temperature during the day. Following incubation, a second rinsing procedure was performed (four washes in PBT over the course of 75 minutes) before mounting the wings on slides with Vectashield, as above.

Wing hinge images

For wing hinge images, a full adult fly was hemisected. First, flies were killed by freezing flies briefly on ice, then dipping in 95% ethanol. Next they were frozen in Tissue-Tek O.C.T. Compound on dry ice for ~3 minutes. Flies were then sliced along the anterior-posterior axis with a razor blade and transferred to a series of wells of ~3mL 4% paraformaldehyde PBS solution until the O.C.T. melted away. Hemisected flies were then transferred to a 0.6 mL tube with fresh fixative for 45 minutes before following the same washing procedure detailed above. Instead of Vectashield, hemisected flies were mounted using the FocusClear-MountClear system (CelExplorer FC-101 and MC-301).

Confocal Imaging and image post-processing

Mounted wings and wing hinges were imaged on a Confocal Olympus FV1000. Images were processed in FIJI (Schindelin et al., 2012 [DOI](#)).

FlyLight confocal stacks

Confocal stacks were downloaded from the gen1 GAL4 and MCFO GAL4 collections on FlyLight (Jenett et al., 2012 [DOI](#); Meissner et al., 2023 [DOI](#)) and displayed as max projections using FIJI. All FlyLight z-stacks for the genotypes in this project are publicly available online at <https://www.janelia.org/project-team/flylight> [DOI](#). For Figures 3 [DOI](#), 5, and 6, VNC expression patterns from the full GAL4 lines were aligned using the Computational Morphometry Toolkit (CMTK) to a female VNC template (Bogovic et al., 2020 [DOI](#)) in FIJI. For Figures 4 [DOI](#) and 7 [DOI](#), MCFO images were used because the full expression patterns were too broad in the whole VNC to visualize the wing sensory neurons in a max projection.

The depth-colored FlyLight MCFO images in Figure 8 [DOI](#) were adjusted to visually highlight single neurons. First, we duplicated the max-projection z-stack and increased the contrast on one copy. Next, we traced the relevant neuron in the original and used this shape to mask the high contrast copy. We then overlaid this masked image onto the original. This method allowed us to highlight single neurons visually in busy MCFO images. Full z-stacks are available through FlyLight.

Peripheral Identification

See Table 2 [DOI](#) for a list of references we used to identify peripheral structures along the wing and near the wing hinge (Cole and Palka, 1982 [DOI](#); Dinges et al., 2021 [DOI](#); Fudalewicz-Niemczyk, 1963 [DOI](#); Hartenstein and Posakony, 1989 [DOI](#); Hertweck, 1931 [DOI](#)). Sensory structures were identified from confocal image stacks by closely scrutinizing the images to see exactly where GFP-labeled neurons were in relation to landmarks such as wing veins and sclerites. Campaniform sensilla were the most straightforward sensory structures to identify thanks to a comprehensive atlas (Dinges et al., 2021 [DOI](#)). The chordotonal organs were identified by their actin-rich attachment cells labeled by phalloidin. The structure on the tegula was identified as a HP due to the appearance of the hairs.

Supplementary information

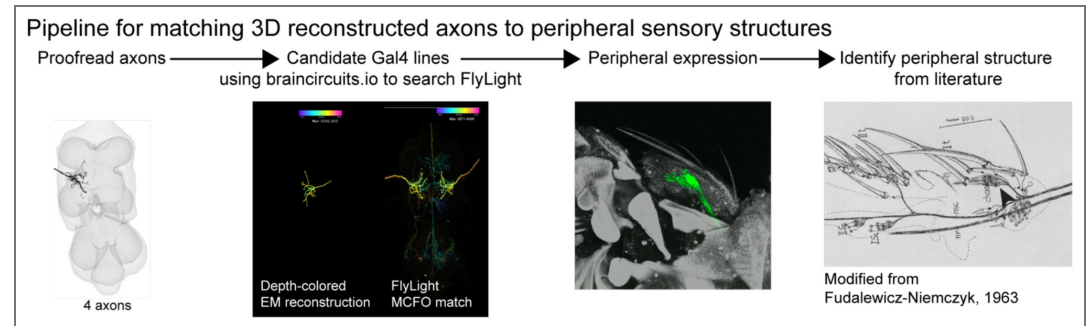


Figure 1 supplement. Pipeline for matching 3D reconstructed axons to sensory neurons on the wing and wing hinge. Driver lines were identified by their sparse expression in the wing nerve, then crossed to a membrane-bound GFP (UAS-mCD8::GFP), imaged with a confocal microscope, and then compared with literature for identification. For details on matching peripheral wing anatomy to literature, see [Table 2](#)

Data availability

Materials availability The genetic driver lines used in this study are listed in Table 3 and are available from the Bloomington Drosophila Stock center. Data and Code availability VNC images are publicly available via FlyLight (<https://www.janelia.org/project-team/flylight>). Confocal stacks of the genetic expression in the wing for each driver line are available for download from Dryad (<https://doi.org/10.5061/dryad.mgqnk99b5>). An annotation table that includes the FANC cell ID and peripheral identification of each segment in detailed is available to the FANC community, as well as on Dryad as a CSV. Analyses and a connectivity table are stored at https://github.com/EllenLesser/Lesser_eLife_2025.

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References

Ammer G., Vieira R.M., Fendl S., Borst A (2022) Anatomical distribution and functional roles of electrical synapses in *Drosophila*. *Curr. Biol* **32**:2022-2036.e4. <https://doi.org/10.1016/j.cub.2022.03.040>

Azevedo A., Lesser E., Phelps J.S., Mark B., Elabbady L., Kuroda S., Sustar A., Moussa A., Khandelwal A., Dallmann C.J., *et al.* (2024) Connectomic reconstruction of a female *Drosophila* ventral nerve cord. *Nature* **631**:360-368 <https://doi.org/10.1038/s41586-024-07389-x>

- Bogovic J.A.**, Otsuna H., Heinrich L., Ito M., Jeter J., Meissner G., Nern A., Colonell J., Malkesman O., Ito K., *et al.* (2020) An unbiased template of the *Drosophila* brain and ventral nerve cord. *PLOS One* **15**:e0236495 <https://doi.org/10.1371/journal.pone.0236495>
- Burt R.**, Palka J (1982) The central projections of mesothoracic sensory neurons in wild-type *Drosophila* and *bithorax* mutants. *Dev. Biol* **90**:99-109 [https://doi.org/10.1016/0012-1606\(82\)90215-9](https://doi.org/10.1016/0012-1606(82)90215-9)
- Büschges A.**, Ramirez J.M., Driesang R., Pearson K.G (1992) Connections of the forewing tegulae in the locust flight system and their modification following partial deafferentation. *J. Neurobiol* **23**:44-60 <https://doi.org/10.1002/neu.480230106>
- Chan W.P.**, Dickinson M.H (1996) Position-specific central projections of mechanosensory neurons on the haltere of the blow fly, *Calliphora vicina*. *J. Comp. Neurol* **369**:405-418 [https://doi.org/10.1002/\(SICI\)1096-9861\(19960603\)369:3<405::AID-CNE6>3.0.CO;2-9](https://doi.org/10.1002/(SICI)1096-9861(19960603)369:3<405::AID-CNE6>3.0.CO;2-9)
- Chapman K.M.**, Duckrow R.B., Moran D.T (1973) Form and Role of Deformation in Excitation of an Insect Mechanoreceptor. *Nature* **244**:453-454 <https://doi.org/10.1038/244453a0>
- Cole E.S.**, Palka J (1982) The pattern of campaniform sensilla on the wing and haltere of *Drosophila melanogaster* and several of its homeotic mutants. *Development* **71**:41-61 <https://doi.org/10.1242/dev.71.1.41>
- Dallmann C.J.**, Luo Y., Agrawal S., Chou G.M., Cook A., Brunton B.W., Tuthill J.C (2024) Presynaptic inhibition selectively suppresses leg proprioception in behaving *Drosophila*. *bioRxiv* <https://doi.org/10.1101/2023.10.20.563322>
- Deora T.**, Singh A.K., Sane S.P (2015) Biomechanical basis of wing and haltere coordination in flies. *Proc. Natl. Acad. Sci. U. S. A* **112**:1481-1486 <https://doi.org/10.1073/pnas.1412279112>
- Dhawan S.**, Huang Z., Dickerson B.H. (2025) Neural connectivity of a computational map for fly flight control. *bioRxiv* <https://doi.org/10.1101/2025.05.29.656834>
- Dickerson B.H.** (2020) Timing precision in fly flight control: integrating mechanosensory input with muscle physiology. *Proc. R. Soc. B Biol. Sci* **287**:20201774 <https://doi.org/10.1098/rspb.2020.1774>
- Dickerson B.H.**, Fox J.L., Sponberg S (2021) Functional diversity from generic encoding in insect campaniform sensilla. *Curr. Opin. Physiol* **19**:194-203 <https://doi.org/10.1016/j.cophys.2020.11.004>
- Dickinson M.H.**, Lehmann F.O., Götz K.G (1993) The active control of wing rotation by *Drosophila*. *J. Exp. Biol* **182**:173-189 <https://doi.org/10.1242/jeb.182.1.173>
- Dickinson M.H.**, Palka J (1987) Physiological properties, time of development, and central projection are correlated in the wing mechanoreceptors of *Drosophila*. *J. Neurosci* **7**:4201-4208 <https://doi.org/10.1523/JNEUROSCI.07-12-04201.1987>
- Dickinson M.H.**, Tu M.S (1997) The Function of Dipteran Flight Muscle. *Comp. Biochem. Physiol. A Physiol* **116**:223-238 [https://doi.org/10.1016/S0300-9629\(96\)00162-4](https://doi.org/10.1016/S0300-9629(96)00162-4)
- Dinges G.F.**, Chockley A.S., Bockemühl T., Ito K., Blanke A., Büschges A (2021) Location and arrangement of campaniform sensilla in *Drosophila melanogaster*. *J. Comp. Neurol* **529**:905-925 <https://doi.org/10.1002/cne.24987>
- Dorkenwald S.**, Schneider-Mizell C.M., Brittain D., Halageri A., Jordan C., Kemnitz N., Castro M.A., Silversmith W., Maitin-Shephard J., Troidl J., *et al.* (2023) CAVE: Connectome Annotation Versioning Engine. *bioRxiv* <https://doi.org/10.1101/2023.07.26.550598>
- Edwards J.S.**, Milner M.J., Chen S.W (1978) Integument and sensory nerve differentiation of *Drosophila* leg and wing imaginal discs in vitro. *Wilhelm Roux Arch. Dev. Biol* **185**:59-77 <https://doi.org/10.1007/BF00848215>
- Fayyazuddin A.**, Dickinson M.H (1999) Convergent mechanosensory input structures the firing phase of a steering motor neuron in the blowfly, *Calliphora*. *J. Neurophysiol* **82**:1916-1926 <https://doi.org/10.1152/jn.1999.82.4.1916>
- Fayyazuddin A.**, Dickinson M.H (1996) Haltere Afferents Provide Direct, Electrotonic Input to a Steering Motor Neuron in the Blowfly, *Calliphora*. *J. Neurosci* **16**:5225-5232 <https://doi.org/10.1523/JNEUROSCI.16-16-05225.1996>

- Field L.H., Matheson T. (1998) Chordotonal Organs of Insects. In: Evans P.D. (Ed). *Advances in Insect Physiology* Academic Press. pp. 1-228 [https://doi.org/10.1016/S0065-2806\(08\)60013-2](https://doi.org/10.1016/S0065-2806(08)60013-2)
- Fudalewicz-Niemczyk Władysława (1963) L'innervation et les organes sensoriels des ailes des Diptères et comparaison avec l'innervation des ailes d'insectes d'autres ordres. *Acta Zoologica Cracoviensis*
- Galili D.S., Jefferis G.S.X.E., Costa M (2022) Connectomics and the neural basis of behaviour. *Curr. Opin. Insect Sci* **100968** <https://doi.org/10.1016/j.cois.2022.100968>
- Gau J., Lynch J., Aiello B., Wold E., Gravish N., Sponberg S (2023) Bridging two insect flight modes in evolution, physiology and robophysics. *Nature* **622**:767-774 <https://doi.org/10.1038/s41586-023-06606-3>
- Gettrup E (1962) Thoracic Proprioceptors in the Flight System of Locusts. *Nature* **193**:498-499 <https://doi.org/10.1038/193498a0>
- Ghyssen A (1980) The projection of sensory neurons in the central nervous system of *Drosophila*: Choice of the appropriate pathway. *Dev. Biol* **78**:521-541 [https://doi.org/10.1016/0012-1606\(80\)90351-6](https://doi.org/10.1016/0012-1606(80)90351-6)
- Ghyssen A (1978) Sensory neurones recognise defined pathways in *Drosophila* central nervous system. *Nature* **274**:869-872 <https://doi.org/10.1038/274869a0>
- Hampel S., McKellar C.E., Simpson J.H., Seeds A.M (2017) Simultaneous activation of parallel sensory pathways promotes a grooming sequence in *Drosophila*. *eLife* **6**:e28804 <https://doi.org/10.7554/eLife.28804>
- Hartenstein V., Posakony J.W (1989) Development of adult sensilla on the wing and notum of *Drosophila melanogaster*. *Development* **107**:389-405 <https://doi.org/10.1242/dev.107.2.389>
- Heide G (1983) Neural mechanisms of flight control in Diptera. *BIONA-Rep* **2**:35-52
- Heide G., Götz K.G (1996) Optomotor control of course and altitude in *Drosophila melanogaster* is correlated with distinct activities of at least three pairs of flight steering muscles. *J. Exp. Biol* **199**:1711-1726 <https://doi.org/10.1242/jeb.199.8.1711>
- Hertweck H. (1931) *Anatomie und Variabilität des Nervensystems und der Sinnesorgane von Drosophila melanogaster (Meigen)* Akademische Verlagsgesellschaft m. b. H, Leipzig.
- Jenett A., Rubin G.M., Ngo T.-T.B., Shepherd D., Murphy C., Dionne H., Pfeiffer B.D., Cavallaro A., Hall D., Jeter J., et al. (2012) A GAL4-driver line resource for *Drosophila* neurobiology. *Cell Rep* **2**:991-1001 <https://doi.org/10.1016/j.celrep.2012.09.011>
- Kays I., Cvetkovska V., Chen B.E (2014) Structural and functional analysis of single neurons to correlate synaptic connectivity with grooming behavior. *Nat. Protoc* **9**:1-10 <https://doi.org/10.1038/nprot.2013.157>
- Koh T.-W., He Z., Gorur-Shandilya S., Menuz K., Larter N.K., Stewart S., Carlson J.R (2014) The *Drosophila* IR20a Clade of Ionotropic Receptors Are Candidate Taste and Pheromone Receptors. *Neuron* **83**:850-865 <https://doi.org/10.1016/j.neuron.2014.07.012>
- Kwon Y., Shen W.L., Shim H.-S., Montell C (2010) Fine Thermotactic Discrimination between the Optimal and Slightly Cooler Temperatures via a TRPV Channel in Chordotonal Neurons. *J. Neurosci* **30**:10465-10471 <https://doi.org/10.1523/JNEUROSCI.1631-10.2010>
- Lee S.-Y.J., Dallmann C.J., Cook A., Tuthill J.C., Agrawal S (2025) Divergent neural circuits for proprioceptive and exteroceptive sensing of the *Drosophila* leg. *Nat. Commun* **16**:4105 <https://doi.org/10.1038/s41467-025-59302-3>
- Lesser E., Azevedo A.W., Phelps J.S., Elabbady L., Cook A., Syed D.S., Mark B., Kuroda S., Sustar A., Moussa A., et al. (2024) Synaptic architecture of leg and wing premotor control networks in *Drosophila*. *Nature* **631**:369-377 <https://doi.org/10.1038/s41586-024-07600-z>
- Li J., Zhang W., Guo Z., Wu S., Jan L.Y., Jan Y.-N (2016) A Defensive Kicking Behavior in Response to Mechanical Stimuli Mediated by *Drosophila* Wing Margin Bristles. *J. Neurosci* **36**:11275-11282 <https://doi.org/10.1523/JNEUROSCI.1416-16.2016>

- Lu B., LaMora A., Sun Y., Welsh M.J., Ben-Shahar Y (2012) ppk23-Dependent Chemosensory Functions Contribute to Courtship Behavior in *Drosophila melanogaster*. *PLoS Genet* **8**:e1002587 <https://doi.org/10.1371/journal.pgen.1002587>
- Marin E.C., Morris B.J., Stürner T., Champion A.S., Krzeminski D., Badalamente G., Gkantia M., Dunne C.R., Eichler K., Takemura S., et al. (2024) Systematic annotation of a complete adult male *Drosophila* nerve cord connectome reveals principles of functional organisation. *eLife* **13** <https://doi.org/10.7554/eLife.97766.1>
- Meinertzhagen I.A., Takemura S., Meinertzhagen I.A., Takemura S., Lu Z., Huang S., Gao S., Ting C.-Y., Lee C.-H (2009) From Form to Function: the Ways to Know a Neuron. *J. Neurogenet* **23**:68-77 <https://doi.org/10.1080/01677060802610604>
- Meissner G.W., Nern A., Dorman Z., DePasquale G.M., Forster K., Gibney T., Hausenfluck J.H., He Y., Iyer N.A., Jeter J., et al. (2023) A searchable image resource of *Drosophila* GAL4 driver expression patterns with single neuron resolution. *eLife* **12**:e80660 <https://doi.org/10.7554/eLife.80660>
- Miyan J.A., Ewing A.W (1984) A wing synchronous receptor for the Dipteran flight motor. *J. Insect Physiol* **30**:567-574 [https://doi.org/10.1016/0022-1910\(84\)90085-4](https://doi.org/10.1016/0022-1910(84)90085-4)
- Moran D.T., Chapman K.M., Ellis R.A (1971) The fine structure of cockroach campaniform sensilla. *J. Cell Biol* **48**:155-173 <https://doi.org/10.1083/jcb.48.1.155>
- Palka J., Lawrence P.A., Hart H.S (1979) Neural projection patterns from homeotic tissue of *Drosophila* studied in *bithorax* mutants and mosaics. *Dev. Biol* **69**:549-575 [https://doi.org/10.1016/0012-1606\(79\)90311-7](https://doi.org/10.1016/0012-1606(79)90311-7)
- Palka J., Malone M.A., Ellison R.L., Wigston D.J (1986) Central projections of identified *Drosophila* sensory neurons in relation to their time of development. *J. Neurosci* **6**:1822-1830 <https://doi.org/10.1523/JNEUROSCI.06-06-01822.1986>
- Pedregosa F., Varoquaux G., Gramfort A., Michel V., Thirion B., Grisel O., Blondel M., Prettenhofer P., Weiss R., Dubourg V., et al. (2011) Scikit-learn: Machine Learning in Python. *J. Mach. Learn. Res* **12**:2825-2830
- Phelps J.S., Hildebrand D.G.C., Graham B.J., Kuan A.T., Thomas L.A., Nguyen T.M., Buhmann J., Azevedo A.W., Sustar A., Agrawal S., et al. (2021) Reconstruction of motor control circuits in adult *Drosophila* using automated transmission electron microscopy. *Cell* **184**:759-774.e18. <https://doi.org/10.1016/j.cell.2020.12.013>
- Pratt B.G., Dallmann C.J., Chou G.M., Siwanowicz I., Walling-Bell S., Cook A., Sustar A., Azevedo A., Tuthill J.C (2025) Proprioceptive limit detectors mediate sensorimotor control of the *Drosophila* leg. *bioRxiv* <https://doi.org/10.1101/2025.05.15.654260>
- Pringle J.W.S (1957) *Insect Flight* Cambridge University Press.
- Pringle J.W.S (1938a) Proprioception In Insects: II. The Action Of The Campaniform Sensilla On The Legs. *J. Exp. Biol* **15**:114-131 <https://doi.org/10.1242/jeb.15.1.114>
- Pringle J.W.S (1938b) Proprioception in Insects: III. The Function of the Hair Sensilla at the Joints. *J. Exp. Biol* **15**:467-473 <https://doi.org/10.1242/jeb.15.4.467>
- Sager G., Pallasdies F., Gowers R., Ravikumar S., Wu E., Colón-Ramos D., Schreiber S., Clark D.A. (2024) Spatial and morphological organization of mitochondria across a connectome. *bioRxiv* <https://doi.org/10.1101/2024.08.21.608912>
- Schindelin J., Arganda-Carreras I., Frise E., Kaynig V., Longair M., Pietzsch T., Preibisch S., Rueden C., Saalfeld S., Schmid B., et al. (2012) Fiji: an open-source platform for biological-image analysis. *Nat. Methods* <https://doi.org/10.1038/nmeth.2019>
- Stocker R.F (1994) The organization of the chemosensory system in *Drosophila melanogaster*: a review. *Cell Tissue Res* **275**:3-26 <https://doi.org/10.1007/BF00305372>
- Thistle R., Cameron P., Ghorayshi A., Dennison L., Scott K (2012) Contact Chemoreceptors Mediate Male-Male Repulsion and Male-Female Attraction during *Drosophila* Courtship. *Cell* **149**:1140-1151 <https://doi.org/10.1016/j.cell.2012.03.045>

- Trimarchi J.R., Jin P., Murphey R.K. (1999) Controlling the Motor Neuron. In: Bradley Ronald J., R.A.H., P.J. (Eds). *International Review of Neurobiology, Neuromuscular Junctions in Drosophila* Academic Press. [https://doi.org/10.1016/S0074-7742\(08\)60548-6](https://doi.org/10.1016/S0074-7742(08)60548-6)
- Trimarchi J.R., Murphey R.K (1997) The shaking-B2 Mutation Disrupts Electrical Synapses in a Flight Circuit in Adult *Drosophila*. *J. Neurosci* **17**:4700-4710 <https://doi.org/10.1523/JNEUROSCI.17-12-04700.1997>
- Virtanen P., Gommers R., Oliphant T.E., Haberland M., Reddy T., Cournapeau D., Burovski E., Peterson P., Weckesser W., Bright J., et al. (2020) SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* **17**:261-272 <https://doi.org/10.1038/s41592-019-0686-2>
- Weber A.I., Daniel T.L., Brunton B.W (2021) Wing structure and neural encoding jointly determine sensing strategies in insect flight. *PLOS Comput. Biol* **17**:e1009195 <https://doi.org/10.1371/journal.pcbi.1009195>
- Whitehead S.C., Leone S., Lindsay T., Meiselman M.R., Cowan N.J., Dickinson M.H., Yapici N., Stern D.L., Shirangi T., Cohen I (2022) Neuromuscular embodiment of feedback control elements in *Drosophila* flight. *Sci. Adv* **8**:eabo7461 <https://doi.org/10.1126/sciadv.abo7461>
- Whitlock K.E., Palka J (1995) Development of wing sensory axons in the central nervous system of *Drosophila* during metamorphosis. *J. Neurobiol* **26**:189-204 <https://doi.org/10.1002/neu.480260204>
- Wilson D.M., Gettrup E (1963) A Stretch Reflex Controlling Wingbeat Frequency in Grasshoppers. *J. Exp. Biol* **40**:171-185 <https://doi.org/10.1242/jeb.40.1.171>
- Wolf H (1993) The Locust Tegula: Significance for Flight Rhythm Generation, Wing Movement Control and Aerodynamic Force Production. *J. Exp. Biol* **182**:229-253 <https://doi.org/10.1242/jeb.182.1.229>
- Yasuyama K., Salvaterra P.M (1999) Localization of choline acetyltransferase-expressing neurons in *Drosophila* nervous system. *Microsc. Res. Tech* **45**:65-79 [https://doi.org/10.1002/\(SICI\)1097-0029\(19990415\)45:2<65::AID-JEMT2>3.0.CO;2-0](https://doi.org/10.1002/(SICI)1097-0029(19990415)45:2<65::AID-JEMT2>3.0.CO;2-0)

Peer reviews

Reviewer #1 (Public review):

Summary:

Lesser et al provide a comprehensive description of *Drosophila* wing proprioceptive sensory neurons at the electron microscopy resolution. This "tour-de-force", provides a strong foundation for future structural and functional research aimed at understanding wing motor control in *Drosophila* with implications to understanding wing control across other insects.

Strengths:

- (1) Authors leverage previous research that described many of the fly wing proprioceptors, and combine this knowledge with EM connectome data such that they now provide a near-complete morphological description of all wing proprioceptors.
- (2) Authors cleverly leverage genetic tools and EM connectome data to tie the location of proprioceptors on the wings with axonal projections in the connectome. This enables them to both align with previous literature as well as make some novel claims.
- (3) In addition to providing a full description of wing proprioceptors, authors also identified a novel population of sensors on the wing tegula that make direct connections with the B1 wing motor neurons implicating the role of tegula in wing movements that was previously underappreciated.

(4) Despite being the most comprehensive description so far, it is reassuring that authors clearly state the missing elements in the discussion.

Weaknesses:

(1) Authors do their main analysis on data from FANC connectome but provide corresponding IDs for sensory neurons in the MANC connectome. I wonder how the connectivity matrix compares across FANC and MANC if the authors perform similar analysis as they have done in Fig. 2. This could be a valuable addition and potentially also pick up any sexual dimorphism.

(2) Authors speculate about presence of gap junctions based on density of mitochondria. I'm not convinced about this given mitochondrial densities could reflect other things that correlate with energy demands in sub-compartment.

Overall, I consider this an exceptional analysis which will be extremely valuable to the community.

<https://doi.org/10.7554/eLife.107867.2.sa3>

Reviewer #2 (Public review):

Summary:

Lesser et al. present an atlas of *Drosophila* wing sensory neurons. They proofread the axons of all sensory neurons in the wing nerve of an existing electron microscopy dataset, the female adult fly nerve cord (FANC) connectome. These reconstructed sensory axons were linked with light microscopy images of full-scale morphology to identify their origin in the periphery of the wing and encoded sensory modalities. The authors described the morphology and postsynaptic targets of proprioceptive neurons as well as previously unknown sensory neurons.

Strengths:

The authors present a valuable catalogue of wing sensory neurons, including previously undescribed sensory axons in the *Drosophila* wing. By providing both connectivity information with linked genetic drive lines, this research facilitates future work on the wing motor-sensory network and applications relating to *Drosophila* flight. The findings were linked to previous research as well as their putative role in the proprioceptive and nerve cord circuitry, providing testable hypotheses for future studies.

Weaknesses:

With future use as an atlas, it should be noted that the evidence is based on sensory neurons on only one side of the nerve cord. Fruit flies have stereotyped left/right hemispheres in the brain and left/right hemisegments in the nerve cord. Comparison of left and right neurons of the nervous system can give a sense of how robust the morphological and connectivity findings are. Unfortunately, this dataset has damage to the right side, making such comparisons unreliable.

<https://doi.org/10.7554/eLife.107867.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors aim to identify the peripheral end organ origin in the fly's wing of all sensory neurons in the Anterior Dorsal Mesothoracic nerve. They reconstruct the neurons and their downstream partners in an electron microscopy volume of a female ventral nerve cord, analyse the resulting connectome and identify their origin with review of the literature and imaging of genetic driver lines. While some of the neurons were already known through previous work, the authors expand on the identification and create a near complete map of the wing mechanosensory neurons at synapse resolution.

Strengths:

The authors elegantly combine electron microscopy neuron morphology, connectomics and light microscopy methods to bridge the gap between fly wing sensory neuron anatomy and ventral nerve cord morphology. Further, they use EM ultrastructural observations to make predictions on the signaling modality of some of the sensory neurons and thus their function in flight.

The work is as comprehensive as state of the art methods allow to create a near complete map of the wing mechanosensory neurons. This work will be of importance to the field of fly connectomics and modelling of fly behavior as well as a useful resource to the *Drosophila* research community.

Through this comprehensive mapping of neurons to the connectome the authors create a lot of hypotheses on neuronal function partially already confirmed with the literature and partially to be tested in the future. The authors achieved their aim of mapping the periphery of the fly's wing to axonal projections in the ventral nerve cord, beautifully laying out their results to support their mapping.

The authors identify the neurons in a previously published connectome of a male fly ventral nerve cord to enable cross-individual analysis of connections and find no indication of sexual dimorphism at the sensory neuron level. Further, together with their companion paper Dhawan et al., 2025 describing the haltere sensory neurons in the same EM dataset, they cover the entire mechanosensory space involved in *Drosophila* flight.

<https://doi.org/10.7554/eLife.107867.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Lesser et al provide a comprehensive description of Drosophila wing proprioceptive sensory neurons at the electron microscopy resolution. This “tour-de-force” provides a strong foundation for future structural and functional research aimed at understanding wing motor control in Drosophila with implications for understanding wing control across other insects.

Strengths:

(1) The authors leverage previous research that described many of the fly wing proprioceptors, and combine this knowledge with EM connectome data such that they now provide a near-complete morphological description of all wing proprioceptors.

(2) The authors cleverly leverage genetic tools and EM connectome data to tie the location of proprioceptors on the wings with axonal projections in the connectome. This enables them to both align with previous literature as well as make some novel claims.

(3) In addition to providing a full description of wing proprioceptors, the authors also identified a novel population of sensors on the wing tegula that make direct connections with the B1 wing motor neurons, implicating the role of the tegula in wing movements that was previously underappreciated.

(4) Despite being the most comprehensive description so far, it is reassuring that the authors clearly state the missing elements in the discussion.

Weaknesses:

(1) The authors do their main analysis on data from the FANC connectome but provide corresponding IDs for sensory neurons in the MANC connectome. I wonder how the connectivity matrix compares across FANC and MANC if the authors perform a similar analysis to the one they have done in Figure 2. This could be a valuable addition and potentially also pick up any sexual dimorphism.

We agree that systematic comparisons will provide valuable insights as more connectome datasets become available. However, the primary goal of this study was to link central axon morphology with peripheral structures in the wing. We deliberately omitted more detailed and quantitative analyses of the downstream VNC circuitry, apart from providing a global view of the connectivity matrix and using it to cluster the sensory axon types. A more detailed and systematic comparison of wing sensorimotor circuit connectivity across different connectome datasets (FANC, MANC, BANC, IMAC) is the subject of ongoing work in our lab, which we feel is beyond the scope of this study. Here, we chose to match the wing proprioceptors to axons in MANC to demonstrate their stereotypy across individuals and to make them more accessible to other researchers. We found no obvious sexual dimorphism at the level of wing sensory neurons. We now note this in the Discussion.

(2) The authors speculate about the presence of gap junctions based on the density of mitochondria. I'm not convinced about this, given that mitochondrial densities could reflect other things that correlate with energy demands in sub-compartments.

We have moved speculation about mitochondria and gap junctions to the Discussion.

(3) I'm intrigued by how the tegula CO is negative for iav. I wonder if authors tried other CO labeling genes like *nompC*. And what does this mean for the nature of this CO. Some more discussion on this anomaly would be helpful.

Based on this suggestion, we have added an image showing that tegula CO neurons are labeled by *nompC-Gal4*.

(4) The authors conclude there are no proprioceptive neurons in sclerite pterale C based on *Chat-Gal4* expression analysis. It would be much more rigorous if authors also tried a pan-neuronal driver like *nsyb/elav* or other neurotransmitter drivers (*Vglut*, *GAD*, etc) to really rule this out. (I hope I didn't miss this somewhere.)

To address this, we imaged OK371-GFP, which labels glutamatergic neurons, in the wing and wing hinge. We saw expression in the wing, as others have reported (Neukomm et al., 2014), but we saw no expression at the wing hinge. Apart from a handful of glutamatergic gustatory neurons in the leg, we are not aware of any other sensory neurons in the fly that are not labeled by *Chat-Gal4*.

Overall, I consider this an exceptional analysis that will be extremely valuable to the community.

We sincerely appreciate the reviewer's positive feedback.

Reviewer #2 (Public review):

Summary:

Lesser et al. present an atlas of Drosophila wing sensory neurons. They proofread the axons of all sensory neurons in the wing nerve of an existing electron microscopy dataset, the female adult fly nerve cord (FANC) connectome. These reconstructed sensory axons were linked with light microscopy images of full-scale morphology to identify their origin in the periphery of the wing and encoded sensory modalities. The authors described the morphology and postsynaptic targets of proprioceptive neurons as well as previously unknown sensory neurons.

Strengths:

The authors present a valuable catalogue of wing sensory neurons, including previously undescribed sensory axons in the Drosophila wing. By providing both connectivity information with linked genetic drive lines, this research facilitates future work on the wing motor-sensory network and applications relating to Drosophila flight. The findings were linked to previous research as well as their putative role in the proprioceptive and nerve cord circuitry, providing testable hypotheses for future studies.

Weaknesses:

(1) With future use as an atlas, it should be noted that the evidence is based on sensory neurons on only one side of the nerve cord. Fruit flies have stereotyped left/right hemispheres in the brain and left/right hemisegments in the nerve cord. The comparison of left and right neurons of the nervous system can give a sense of how robust the morphological and connectivity findings are. Here, the authors have not compared the left and right side sensory axons from the wing nerve, leaving potential for developmental variability across samples and left/right hemisegments.

The right ADMN nerve in the FANC dataset is partially severed, making left/right comparisons unreliable (see Azevedo 2024, Extended Data Figure 4). We have updated the text to explain this within the Methods section of the paper.

(2) Not all links between the EM reconstructions and driver lines are convincing. To strengthen these, for all EM-LM matches in Figures 3-7, rotated views of the driver line (matching the rotated EM views) should be shown to provide a clearer comparison of the data. In particular, Figure 3G and Figure 7B are not very convincing based on the images shown. MCFO imaging of the driver lines in Figure 3G and 7B would make this position stronger if a clone that matches the EM reconstruction could be identified.

Many of the z-stack images in the paper are from the Janelia FlyLight collection, and unfortunately their imaging parameters were not optimized for orthogonal views. Rotated views are blurry and not especially helpful for comparison to EM reconstruction. We now point out in the text that interested readers can access the z-stacks from FlyLight to see the dorsal-ventral projections.

Regarding Figure 3G and 7B, we have added markers to the image with corresponding descriptions in the legend to guide the reader through the image of the busy driver line. Although these lines label many cells in the VNC as a whole, they sparsely label cells in the ADMN, making them nonetheless useful for identifying peripheral sensory neurons.

(3) Figure 7B looks like the driver line might have stochastic expression in the sensory neuron, which further reduces confidence in the result shown in Figure 7C. Is this expression pattern in the wing consistently seen? Many split-GAL4s have stochastic expressions. The evidence would be strengthened if the authors presented multiple examples (~4-5) of each driver line's expression pattern in the supplement.

Figure 7B shows sparse labeling of the driver line using the MCFO technique, as specified in the legend. Its unilateral expression is therefore not due to stochastic expression of the Gal4 line. We have added the “MFCO” label to the image to clarify.

(4) Certain claims in this work lack quantitative evidence. On line 128, for instance, “Overall, our comprehensive reconstruction revealed many morphological subgroups with overlapping postsynaptic partners, suggesting a high degree of integration within wing sensorimotor circuits.” If a claim of subgroups having shared postsynaptic partners is being made, there should have been quantitative evidence. For example, cosine similarity amongst members of each group compared to the cosine similarity of shuffled/randomised sets of axons from different groups. The heat map of cosine similarity in Figure 2B alone is not sufficient.

We agree that illustrating the extent of shared postsynaptic partners across subgroups strengthens this point. We added a visualization showing pairwise similarity scores for within- and between-cluster neuron pairs (Figure 2B inset). We also performed a permutation test to determine that within-cluster similarity is significantly higher than between clusters, and we report the test in the results as well as the figure legend. This analysis provides a more quantitative summary of the qualitative trends in connectivity that are summarized in Figure 2B.

(5) Similarly, claims about putative electrical connections to b1 motor neurons are very speculative. The authors state that “their terminals contain very densely packed mitochondria compared to other cells”, without providing a quantitative comparison to other sensory axons. There is also no quantitative comparison to the one example of another putative electrical connection from the literature. Further, it should be noted that this connection from Trimarchi and Murphey, 1997, is also stated as putative on line 167, which further weakens this evidence. Quantification would strongly strengthen this position. Identification of an example of high mitochondrial density at a confirmed electrical connection would be even better. In the related discussion section “A potential metabolic specialization for flight circuitry”, it should be more clearly noted that the dense mitochondria could be unrelated to a putative electrical connection. If the authors have an alternative hypothesis about the mitochondria density, this should be stated as well.

We agree with the reviewer that the link between mitochondrial density and metabolic specialization is purely speculative in this context. Based on reviewer feedback, we have moved all mention of the relationship between mitochondrial density and gap junction coupling to the Discussion. We acknowledge that this may seem like a somewhat random and not quantitatively supported observation. However, we found the coincidence striking and worthy of mention, though it is only tangentially relevant to the rest of the paper. From conversations with colleagues, we have also heard that this relationship is consistent with as yet unpublished work in other model organisms (e.g., zebrafish, mouse).

The electrical coupling to b1 motor neurons is well-established (Fayyazuddin and Dickinson, 1999), and we have updated the text to state this more clearly. However, we agree that whether the specific neurons we have identified based on their anatomy are the same ones functionally identified through whole-nerve recordings remains unknown.

(6) It would be appropriate to cite previous work using a similar strategy to match sensory axons to their cell bodies/dendrites at the periphery using driver lines and connectomics (see Figure 5 for example in the following paper: <https://doi.org/10.7554/eLife.40247>).

At this point, there are now dozens of papers that match the axons of sensory neurons to their cell bodies/dendrites in the periphery by comparing light microscopy and connectomics. When we dug in, we found examples in *C. elegans*, *Ciona intestinalis*, zebrafish, and mouse, all published prior to the study cited above. For basically every animal for which scientists have acquired EM volumes of neural tissue, they have used other anatomical labeling methods to determine cell types inside and outside the imaged volume. In summary, we found it difficult to establish a single primary citation for this approach. In lieu of this, we have added a citation to an earlier review by a pioneer in EM connectomics that discusses the general approach of matching cells across different labeling/imaging modalities (Meinertzhagen et al., 2009).

The methods section is very sparse. For the sake of replicability, all sections should be expanded upon.

We have expanded the methods section, and also a STAR methods table.

Reviewer #3 (Public review):

Summary:

The authors aim to identify the peripheral end-organ origin in the fly's wing of all sensory neurons in the anterior dorsomedial nerve. They reconstruct the neurons and their downstream partners in an electron microscopy volume of a female ventral nerve cord, analyse the resulting connectome, and identify their origin with a review of the literature and imaging of genetic driver lines. While some of the neurons were already known through previous work, the authors expand on the identification and create a near-complete map of the wing mechanosensory neurons at synapse resolution.

Strengths:

The authors elegantly combine electron microscopy, neuron morphology, connectomics, and light microscopy methods to bridge the gap between fly wing sensory neuron anatomy and ventral nerve cord morphology. Further, they use EM ultrastructural observations to make predictions on the signaling modality of some of the sensory neurons and thus their function in flight.

*The work is as comprehensive as state-of-the-art methods allow to create a near-complete map of the wing mechanosensory neurons. This work will be of importance to the field of fly connectomics and modelling of fly behavior, as well as a useful resource to the *Drosophila* research community.*

Through this comprehensive mapping of neurons to the connectome, the authors create a lot of hypotheses on neuronal function, partially already confirmed with the literature and partially to be tested in the future. The authors achieved their aim of mapping the periphery of the fly's wing to axonal projections in the ventral nerve cord, beautifully laying out their results to support their mapping.

*The authors identify the neurons in a previously published connectome of a male fly ventral nerve cord to enable cross-individual analysis of connections. Further, together with their companion paper, Dhawan et al. 2025, describing the haltere sensory neurons in the same EM dataset, they cover the entire mechanosensory space involved in *Drosophila* flight.*

Weaknesses:

The connectomic data are only available upon request; the inclusion of a connectivity table of the reconstructed neurons would aid analysis reproducibility and cross-dataset comparisons.

We have added a connectivity table as well as analysis scripts in the github repository for the paper (https://github.com/EllenLesser/Lesser_eLife_2025 [↗](#)).

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

The methods section should be expanded in every aspect. Most pressing sections are:

(1) Data and Code availability: All code should be included as a Zenodo database, the suggestion to ask authors for code upon request is inappropriate.

We have added all code to a public github repository, which is now linked in the Methods section.

(2) Samples: Standard cornmeal and molasses medium should have a reference, as many institutes use different recipes.

The recipe used by the University of Washington fly kitchen is based on the Bloomington standard Cornmeal, Molasses and Yeast Medium recipe, which can be found at <https://bdsc.indiana.edu/information/recipes/molassesfood.html> [↗](#). The UW recipe is slightly modified for different antifungal ingredients and includes tegosept, propionic acid, and phosphoric acid.

(3) Table 3: Driver lines labelling wing sensory neurons: The genetic driver lines should have associated Bloomington stock centre numbers. Additionally, relevant information for effector lines used should be included in the methods.

We now include the Bloomington stock numbers and more information on effector lines in the STAR methods table.

Minor corrections:

(1) Lines 119-120: "Notably, many of the axons do not form crisp cluster boundaries, suggesting that multimodal sensory information is integrated at early stages of sensory processing." We do not follow the logic of this statement and suspect it is a bit too speculative.

We removed this sentence from the manuscript.

(2) Figure 1: The ADMN is missing in the schematics and would be helpful to depict for non-experts. Is this what is highlighted in Figure 1D?

Yes, and we now label 1D as the ADMN wing nerve.

(3) Figure 1B: Which driver lines are being depicted here? Looking at Table 3 does not clarify. It should be specified at least in the figure legend.

As stated in the legend, we include a table of all of the driver lines we screened and which sensory structures they label.

(4) Figure 1C: There are some minor placement issues with the text in the schematic. There is an arrow very close to the "CO" on the top right, which makes the "O" look like

| the symbol for male. “ax ii” is a bit too close to the wing hinge

We updated the figure to address this issue.

| (5) Figure 1D: The outlined grey masks are not clear. The use of colour would be very useful for the reader to help understand what the authors are referring to here

We now use color for the masks.

| (6) Figure 2A: It is unclear if the descending neuron and non-motor efferent neuron are not shown because they are under the described threshold, or to simplify the plot. They should be included in the plot if over the threshold.

We have updated the legend to specify that the exclusion of the descending and non-motor efferent neurons are to visually simplify the plot. We include % of sensory output to each of these neurons in the legend, and they are included in the connectivity matrix data in the public GitHub repository associated with the paper, included in the Methods.

| (7) Figure 2B: What clustering is used specifically? The method says it’s from Scikit-learn, but there are many types of clustering available in this package.

We now include the specific clustering type used in the Methods section, which is agglomerative clustering.

| (8) Figure 3A: What does the green box behind the plot represent?

The green box represents the tegula CO axons, which we now specify in the legend.

| (9) Figure 3C: the “C” is clipped at the top.

We updated the figure to address this issue.

| (10) Figure 4A: the main text says a “group of four axons” (line 203) while the figure says 5 axons.

We updated the text to address this issue.

| (11) Line 360: “We found that the campaniform sensilla on the tegula provide the most direct feedback onto wing steering motor neurons”. We struggled to find where this was directly shown, because several sensory axon types directly synapse onto motor neurons.

We now specify in the text that this finding is shown in Figure 3.

Reviewer #3 (Recommendations for the authors):

I would like to congratulate the authors on their beautiful, easy-to-read, and easy-to-comprehend manuscript, with clear figures and nice visualizations. This work provides a valuable resource that will contribute to the interpretability of connectomic data and further to connectome-based modeling of fly behavior.

We sincerely appreciate the reviewer’s positive feedback.

<https://doi.org/10.7554/eLife.107867.2.sa0>